

Georgia Institute of Technology

PMASE ASE6103 - Capstone

Project Inventory Compliance and Kitting System (PICKS)

Completed for

Calvin Coker
Georgia Tech Research Institute

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Team 2

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Executive Summary

This should be created at the end of the project during ASE 6104, prior to submission.

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List of Acronyms*

Table 1 List of Acronyms

Acronym	Definition
API	Application Programming Interface
BDD	Block Definition Diagram
BOM	Bill of Materials
C3D	Command, Control & Communications Division
CDRL	Contract Data Requirements List
CMMC	Cybersecurity Maturity Model Certification
CMMC 2.0	Cybersecurity Maturity Model Certification 2.0
ERP	Enterprise Resource Planning
GTRI	Georgia Tech Research Institute

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Acronym	Definition
ID	Identification
MES	Manufacturing Execution System
MRP	Material Requirements Planning
NIST	National Institute of Standards and Technology
OV-1	Operational Viewpoint 1
OTS	Off-The-Shelf
PICKS	Project Inventory Compliance and Kitting System
PLM	Product Lifecycle Management
SNR	Stakeholder Need Requirement
SYS	System Requirement
TLS	Transport Layer Security

1. Project Overview*

1.1 Topic Summary

Currently, Georgia Tech Research Institute (GTRI) faces significant inventory tracking challenges due to fragmented infrastructure, inconsistent standards, and decentralized project practices across its eight laboratories. Differences in contractual requirements, lab-specific workflows, and project-specific repositories make system integration and enterprise-wide alignment difficult.

The PICKS capstone project focuses on designing a framework for a desired system that would help unify these disparate processes, enable compliance with emerging requirements, and provide greater transparency into system components and data flows. This system would help GTRI laboratories manage parts, materials, and equipment more efficiently, securely, and transparently across its projects.

The system framework being developed is similar in concept to a existing core business systems (manufacturing execution system (MES), Enterprise resource planning (ERP), and Product life cycle management (PLM)), in that it would track materials from raw components through finished products, monitor assets used throughout workflows, record and manage operational data inputs, trace individual components by batch or configuration, and automate workflow steps such as material handling and process transitions. Rather than building a final software product, the team's role is to define what such a system should do, how it should function, and what requirements it must meet so the organization can later decide whether to develop or acquire a solution. As part of this framework, a list of manufacturing execution system (MES) options and options for roles based access control will be provided. This framework will help GTRI leadership evaluate implementation options—such as internal development, external acquisition, or hybrid approaches—and guide future decision-making.

1.2 Sponsor Information

Sponsor Organization

The project sponsor is Georgia Tech Research Institute, an established applied research organization that supports government and defense sponsors through engineering, technology development, and testing. As GTRI has grown and taken on more complex contracts, it has faced increasing pressure to strengthen supply-chain traceability, documentation, and compliance practices. Existing processes are often fragmented across labs and projects, making it harder to maintain consistency, efficiency, and audit readiness. The proposed system framework aims to address these challenges by providing a unified yet flexible approach to inventory, workflow, and lifecycle tracking.

Sponsor Representatives

Calvin Coker is the PICKS capstone sponsor. In his current role, Calvin leads the Applied Logistics Engineering Support Branch Command, Control & Communications Division (C3D) at GTRI and serves as the primary project liaison. He provides operational insight, defines real-world logistics needs, and ensures the project reflects practical constraints and priorities.

John Davis is a Research Associate in the C3D division at GTRI and has been with the institute for just over two years, working within Cal Coker's department. Prior to joining GTRI, John worked as a Senior Software Consultant in Atlanta, specializing in warehouse management and enterprise logistics systems.

1.3 Capstone Project Team

The PICKS project team is comprised of five Georgia Tech systems engineering graduate students. The team is responsible for analyzing the problem space and designing a structured solution framework.

Their responsibilities include:

- Interviewing stakeholders to understand workflows and challenges
- Identifying operational gaps and risks
- Defining system capabilities and requirements
- Designing high-level architecture concepts
- Demonstrating system behavior through example use cases
- Providing options for manufacturing execution system (MES) for future integration
- Providing roles based access control (RBAC) options for future integration

Team members contribute equally across roles typical of early system development phases, including requirements analysis, architecture design, workflow modeling, integration planning, and technical documentation. Collaboration across these roles ensures that technical feasibility, user needs, and organizational realities remain aligned.

Table 2 Team experience summary

Team Member	Experience
Sasha Rios	Process engineer with over 9 years of experience in new product development. Extensive experience in developing and implementing manufacturing processes including supply chain development and risk assessments for small volume high complexity products. These experiences include workflow and router generation, core system integration (MES, ERP, PLC etc.), BOM setup, etc.
Naksha Roy	Systems Engineer in Aeronautics/Defense industry with a background in Chemical Engineering and Business Management. Experienced in MBSE (Cameo/SysML/3DX), requirements generation and management, modeling and simulation, HSI, trade studies and statistical analysis, automating via Java-based Python scripts and programming tools like MATLAB. Support multiple projects in Mission Systems & SW architecture development and enterprise digital transformation efforts, including a new company-wide Parts Registry tool optimization.
Lauren Campbell	An FFRDC systems analyst and engineer with substantial project leadership experience across several multimillion-dollar advanced R&D programs and prototyping efforts coming from 18 years of combat aviation experience in the US Air Force as an aerial gunner, sensor operator, and air battle manager. Diverse operational experience across 7 Combatant Commands, Joint and Coalition Operations, Contingency Planning and Logistics, and dozens of Live, Virtual, and Constructive Exercises.
Kate Soliven	Full-stack engineer with experience developing real-time AI systems, backend services, and rapid prototypes for government (NASA, Space Force, etc.) and R&D programs. Work spans backend and frontend development including containerization, streaming pipelines, and database modeling. Strong systems engineering background supporting requirements development, trade studies, sensor integration, and modeling and simulation.
Anna Tri	Systems Engineer in Aviation/Defense industry with a background in biomedical engineering. Experience in interpreting customer requirements and translating them into system level and derived requirements and managing them in a requirements database. Experience in model-based systems engineering (MBSE (Cameo/SysML)), human systems integration (HSI), preparing technical design reviews such as systems requirements review, preliminary design review, and critical design reviews, developing test plans, mission and weapons systems, and working directly with stakeholders such as the U.S. government. Holds a Six Sigma Yellow Belt and Project Management Program certification.

1.4 Project Plan

1.4.1. Architecture Refinement & Minimum Viable Product (MVP) Definition

- Finalize the MVP scope with operational, enterprise, and governance stakeholders.
- Prioritize requirements using the weighted scoring model (mission alignment, MVP criticality, complexity penalty) to establish Very High, High, and Medium tiers.
- Mature architecture views for customer feedback: OV 1, functional decomposition, interfaces/draft ICD, and data flows using finalized requirements.
- Validate assumptions about Oracle/Workday integration, legacy lab tools, and compliance boundaries, including data dependencies across Logistics, Program Management, and Lab Systems.
- Confirm feasibility of core architectural elements: BOM infrastructure, router framework, unique ID generation, and cross system synchronization.
- Identify existing and future core systems and boundaries, including ERP, MES candidates, Program Management, and Governance subsystems.
- Define system of systems scope, ensuring PICKS functions as a single integrated enterprise system with centralized governance and shared compliance objectives.

1.4.2. Planning & Early Development

- Develop a prototype plan aligned with MVP workflows: receiving, inventory tracking, kitting, and build execution.
- Align and implement foundational data structures: part numbers, unique IDs, BOM framework, project metadata, and cross system mapping rules.
- Build initial workflow logic for receiving and inventory updates, incorporating Logistics System behaviors (receiving processing time, inventory accuracy).
- Incorporate Program Management System behaviors into planning logic (forecasting, reporting, configuration control).
- Conduct early stakeholder walkthroughs to confirm direction and adjust scope as needed.
- Begin defining integration points for ERP procurement data, Workday financial data, and PEDYN legacy records

1.4.3. Prototype & Integration Modeling (Transition to Post-Capstone)

- Identify MES off the shelf, custom, and open source options with system integration considerations.
- Create RBAC protocol options with evaluation of cost, complexity, and compliance alignment.
- Extend prototype architecture to include kitting and build execution workflows, including traceability chain from components → kits → builds.
- Integrate basic audit logging and access control scaffolding to support governance requirements.
- Generate prototype workflows for component receipt, vendor mapping, part number synchronization, and downstream MES updates.
- Model external system behaviors and interfaces at the architecture level: Oracle ERP, Workday, Program Management System, and Lab System.

- Develop integration V&V protocol plans, including the first three test cases:
 - Test Case 1: ERP Purchase Order Data Exchange
 - Test Case 2: Vendor Data Mapping
 - Test Case 3: Part Number Synchronization
 - Additional Test Cases as identified
- Prepare interim presentation and executive summary for GTRI leadership.

1.4.4. Testing, Evaluation & Architecture Maturation (Post Capstone)

- Down select RBAC and MES options based on system needs, integration constraints, and compliance requirements.
- Execute developed V&V protocols on prototype workflows, including PO exchange, vendor mapping, part number synchronization, and future test cases.
- Conduct usability testing with receiving personnel, lab technicians, and QA stakeholders to validate accessibility and workflow clarity.
- Evaluate error handling behavior for incorrect scans, shortages, mismatches, format errors, duplicate detection, and configuration issues.
- Validate traceability completeness and data propagation across Logistics, Program Management, and Lab Systems.
- Update architecture products based on test results, including interface definitions, data models, workflow logic, and system of systems boundaries.
- Mature governance concepts: auditability, logging, access control, compliance alignment, and integration security.
- Refine system design as technology selections (Oracle ERP variant, MES solution) are finalized and additional requirements are analyzed.

2. Problem Formulation*

1. Address the following regarding the purpose of this system:

a. What is the purpose? What problem must be solved?

The purpose of the proposed system is to improve visibility and control of inventory and kits so that project teams can efficiently manage materials, reduce duplication, and ensure accurate tracking throughout the project lifecycle. As the concept evolves, its role expands beyond basic inventory management to supporting compliance with contract-driven requirements such as CDRLs and DIIDs, enabling enterprise-level governance, and informing future make-versus-buy and acquisition decisions. In the long term, the architectural framework—or a system developed from it—could be reused across multiple labs and programs, providing a scalable solution for broader organizational needs.

b. Can the purpose be identified at the onset, and how can the purpose evolve?

While the core purpose can be identified early as improving operational transparency and control, it is expected to evolve as stakeholder needs, regulations, and organizational priorities change.

- c. **Are there clearly identified customers, clients, and/or stakeholders? If so, please identify them and their particular vested interest. What values, preferences, beliefs, and biases are driving each of them?**

Key stakeholders include GTRI project teams, enterprise leadership, and external government sponsors, each motivated by different priorities such as efficiency, compliance, cost control, and risk reduction.

- d. **In terms of the system's purpose, who benefits, who supplies, who pays, and who loses?**

The primary beneficiaries are project teams, the broader organization, and sponsoring clients who gain improved reliability and traceability. The system would be funded internally by GTRI through overhead or investment resources, with development initially supported by the capstone team and potentially later by internal engineering teams or external vendors. Those who may experience disadvantages include teams accustomed to informal or highly customized processes, legacy tool owners, and users facing short-term productivity impacts during transition and adoption.

2. **Address the following regarding the context of the architecting project:**

- a. **What is the system-of-interest to the architecting effort (at least, the assumed system-of-interest), especially in relation to adjacent systems or system boundaries?**

The system-of-interest for this architecting effort is envisioned as a scalable, commercial-off-the-shelf–based solution designed to improve visibility, traceability, and compliance across organizational inventory and supply chain processes while clearly defining system boundaries with adjacent tools.

- b. **What sort of system does the sponsor believe will eventually emerge?**

The sponsor anticipates a unified platform that addresses growing concerns about supply chain vulnerabilities, material traceability, and cybersecurity compliance requirements such as CMMC 2.0. Although the system is intended to function as a single, broadly applicable solution across labs, it will selectively interface with existing internal tools—for example, referencing procurement data—without tightly integrating with external state-managed systems, which are explicitly out of scope.

- c. **What is the apparent technology level? That is, everything needed to accomplish the basic purpose well within the current state-of-practice, pushing the state-of-practice, or well beyond it?**

From a technology standpoint, the effort is considered to be pushing the current state of practice: current processes track inventory effectively after receipt but lack integration and standardization earlier in the lifecycle, particularly across procurement, manufacturing transitions, and early product management. Desired capabilities include features commonly found in PLM, MES, and ERP Systems, such as BOM and router management, kitting support, lifecycle quality tracking, component genealogy, counterfeit detection, unique identifiers, and

intuitive user interfaces; however, budget limitations and organizational resistance constrain adoption of commercial tools.

d. What hard constraints are believed to exist? Are they really hard constraints or just assumptions?

Several hard constraints shape the effort, including mandatory cybersecurity compliance, a Windows-only operating environment, and state-level procurement limitations, along with nontechnical constraints such as funding limits and concerns about restricting research flexibility.

e. What resources are available for the effort, can more be acquired, if so how, and what are the expectations for interacting with the sponsor? Is the sponsor prepared to engage in value discussions with the team, and is sponsor time available to have the discussions?

The sponsor is willing to actively engage with the team, provide information as allowed, and participate in value discussions, while a parallel working group pursuing an MES request for proposal may inform or benefit from the architecture results.

f. When the architecture effort is complete, what will be done with its products? Will they be used to start a system acquisition, to guide other acquisitions, to guide research and development (R&D) activities, to mark off completion of a bureaucratic requirement, or for some other purpose?

The primary outcome of the architecture effort is to produce a framework-level system architecture that defines what the system should do and how it should operate, along with suggested requirements. This framework will guide make-or-buy decisions, inform future acquisitions, and potentially scale to other GTRI labs. A rough software prototype or a single illustrative use case demonstrating system flow is considered a valuable—but optional—stretch goal. The minimum viable deliverable is the architectural framework itself.

g. Are the purposes of the system-of-interest, the architecting effort, and the architecture documentation to be developed all consistent with each other?

The purposes of the system, architecting effort, and documentation are aligned, all aiming to define and communicate a trusted, compliant, and scalable solution that can serve as a foundation for future development. Ultimately, the investigation is motivated not by a single new technology but by sponsor needs driven by increasing contract complexity, tightening regulatory demands, supply chain risk, and fragmented tooling across labs, with the goal of reducing long-term costs, improving compliance, and enabling more structured and resilient operations.

h. What is motivating the investigation into constructing the system-of-interest? Is it the sponsor needs, a new technology believed to be able to create value, or some other reason?

The motivation for this effort stems from increasing contract complexity, tightening regulations, supply chain risk concerns, and inconsistent tooling across labs. As GTRI's portfolio grows in size and value, the limitations of fragmented, lab-specific solutions have become more pronounced. The sponsor views this effort as a way to reduce long-term costs, improve compliance, and enable more structured, resilient operations using internal funding where possible.

3. Applicable Heuristics*

One key heuristic taken from Maier Appendix A is that *good products are not enough; implementations matter* (Morris & Ferguson, 1993)—emphasizes that successful systems must be introduced in ways that account for human adoption, workflows, and incentives, not just technical performance. In the PICKS project, this principle is reflected in the decision to pursue incremental implementation rather than a single “big bang” deployment, recognizing that a phased framework or demonstrative use case is more realistic and valuable than attempting immediate enterprise-wide rollout. It also appears in the recognition that cross-lab cooperation is essential: without clear workflows, minimal disruption, and stakeholder buy-in, even a technically strong system would fail in practice.

A complementary personal heuristic—that legacy processes are design constraints—further shapes the architecture approach by treating existing lab-specific tools, spreadsheets, and institutional practices not as obstacles to eliminate but as realities the system must accommodate. Together, these heuristics guide the project toward a solution that is technically sound, organizationally adoptable, and realistically implementable.

4. Requirements*

4.1 Stakeholder Needs

To ensure alignment of scope between the capstone team and GTRI project sponsors, five high-priority stakeholder needs were identified based on initial interviews and subsequent clarification discussions. These were reviewed with the sponsors to confirm concurrence and shared understanding. The agreed-upon stakeholder needs are as follows:

- **Stakeholder Need 1: Inventory Visibility and Lifecycle Workflow Support**
This need addresses PICKS' capability to provide real-time visibility of inventory quantities, locations, project ownership, and allocation status while supporting standardized lifecycle workflows such as receiving, inspection, stocking, transfer, and disposition.
- **Stakeholder Need 2: Chain-of-Custody and Traceability for Critical Components**
This need relates to PICKS' ability to maintain end-to-end traceability for critical components,

including supplier, purchase order, inspection, storage, user interaction, and build integration history.

- **Stakeholder Need 3: Kitting and Configuration Management**

This need involves PICKS' ability to enable controlled creation, tracking, and issuance of prototype kits derived from approved Bills of Materials (BOMs) to support configuration integrity and repeatability.

- **Stakeholder Need 4: Compliance and Contract Artifact Support**

This need addresses PICKS' capability to identify contract-driven documentation requirements, link operational records to required artifacts, and generate reports to support audit readiness and sponsor delivery.

- **Stakeholder Need 5: Interface Definitions and Readiness**

This need captures PICKS' ability to define and support structured integration with external enterprise systems and legacy tools to ensure accurate and secure data exchange.

- **Stakeholder Need 6: Enterprise Integration**

This need captures PICKS' ability to integrate with future enterprise systems while addressing gaps in workflow automation, configuration management, traceability, and role-based access control.

4.2 Stakeholder Requirements

Once the stakeholder needs were defined, the team worked the sponsors to generate stakeholder requirements for each need. Figure 1 provides a view of the stakeholder requirements and their traceability to the associated stakeholder need. The requirements analysis decomposed each high-level need into clear, testable stakeholder requirements (SNRs) that will serve as the basis for subsequent derived system requirements. For Stakeholder Need 1, requirements such as SNR-1.1 Tracked Inventory and SNR-1.2 Inventory Lookup are foundational because they establish a single authoritative source of inventory data, directly addressing the sponsor's concern with fragmented spreadsheets and manual tracking. SNR-1.3 Automated Workflows is particularly significant, as configurable lifecycle workflows reduce human error, improve repeatability across labs, and support scalability as project complexity increases. For Stakeholder Need 2, requirements including SNR-2.1 Component Provenance and SNR-2.2 Component History Log are critical from a risk and compliance standpoint, enabling audit readiness, counterfeit mitigation, and informed sourcing decisions. The linkage requirement in SNR-2.4 Critical Components Identification further ensures that traceability is not passive but actively tied to supplier and inspection data, strengthening governance. For Stakeholder Need 3, requirements such as SNR-3.1 Kit Creation Capability and SNR-3.2 Component Availability reinforce configuration control and reduce build delays, while SNR-3.3 Assembly Status Visibility improves coordination between engineering and logistics teams.

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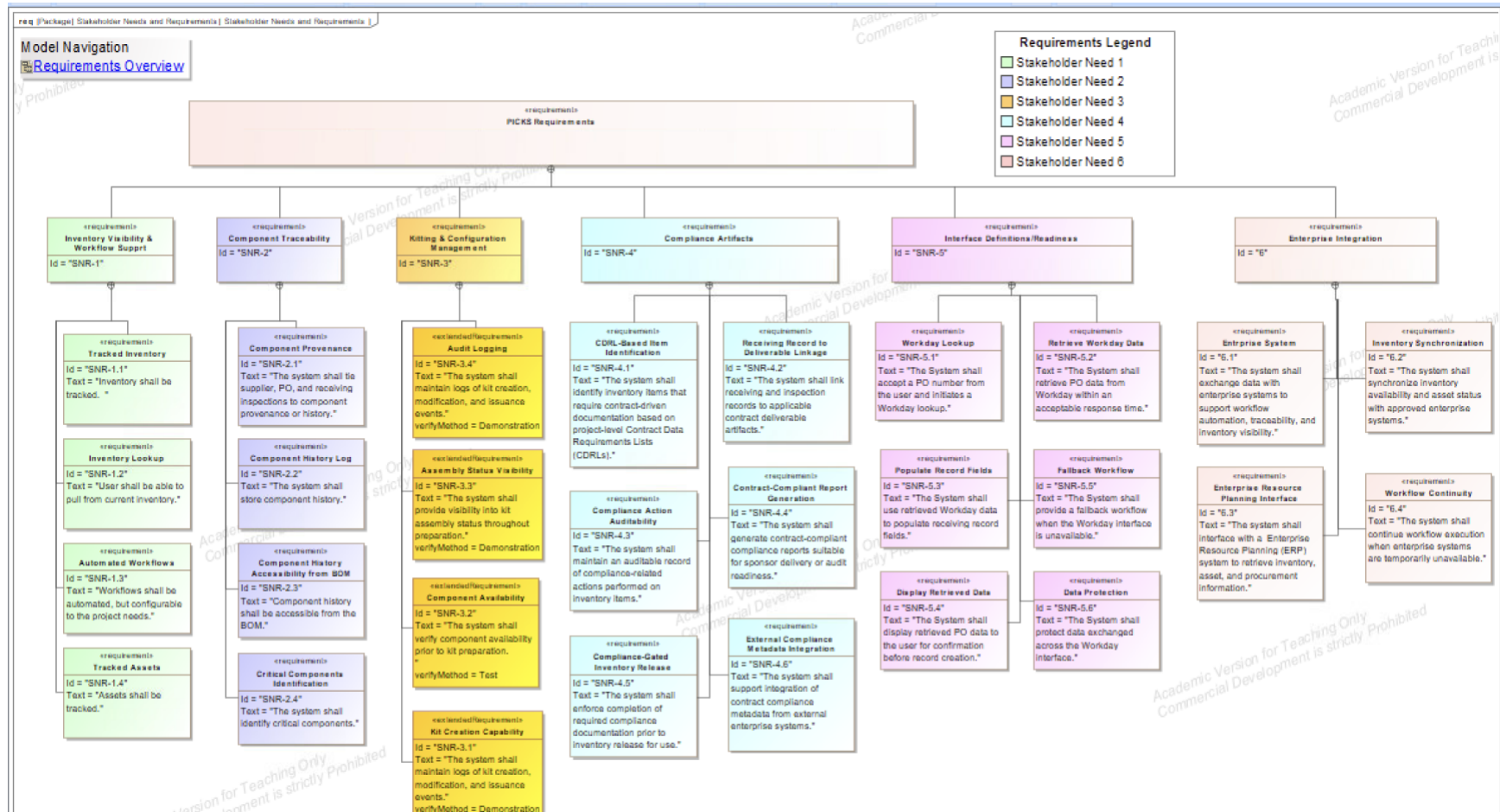


Figure 1. PICKS Stakeholder Needs and Stakeholder Requirements Block Definition Diagram (BDD)

For Stakeholder Needs 4 and 5, the SNRs extend the system from operational inventory management into compliance governance and enterprise integration. Under Need 4, requirements such as SNR-4.1 CDRL-Based Item Identification and SNR-4.2 Receiving Record to Deliverable Linkage ensure that contract-driven documentation requirements are embedded directly within inventory workflows rather than managed externally. SNR-4.3 Compliance Action Auditability introduces a critical governance control by requiring auditable records of compliance-related actions, while SNR-4.4 Contract-Compliant Report Generation and SNR-4.5 Compliance-Gated Inventory Release formalize enforcement mechanisms that prevent release of inventory without required documentation, shifting compliance from passive reporting to active control. SNR-4.6 External Compliance Metadata Integration further reflects architectural considerations by acknowledging the need for structured data exchange with other enterprise systems. Under Need 5, functional requirements such as SNR-5.1 Workday Lookup and SNR-5.3 Populate Record Fields reduce manual entry and improve data accuracy, while non-functional requirements including SNR-5.4 Display Retrieved Data, SNR-5.5 Fallback Workflow, and SNR-5.6 Data Protection introduce usability, reliability, and security expectations. Together, these requirements represent a combination of functional, interface, reliability, and security categories and will drive derived system requirements related to factors like API integration, error handling, encryption, audit logging, and performance thresholds.

Stakeholder Needs 6 is a new addition to our requirements. With the announcement of the State ERP initiative, the team shifted the scope of PICKS from an enterprise inventory management system to a workflow, kitting, and compliance-focused architecture. Rather than duplicating ERP functionality, PICKS is designed to integrate with future enterprise systems while addressing gaps in workflow automation, configuration management, traceability, and role-based access control.

4.3 Derived System Requirements

From each of the SNRs described above, a corresponding set of system-level requirements for PICKS was derived. In the following paragraphs, these system requirements are presented and discussed by stakeholder need to clearly show their decomposition.

The system requirements derived for SNRs related to Need 1 (shown in Figure 2) translate high-level visibility expectations into structured data management and workflow execution behaviors. Foundational requirements such as SYS-1.1.1.1 (Unique ID) and SYS-1.4.1 (Unique ID for Assets) establish persistent identification mechanisms that enable traceability and cross-project inventory control, while core functional requirements like SYS-1.2.1 (Searchable Inventory) and SYS-1.1.1.2 (Inventory Sync upon Issuance) directly support real-time visibility and accurate quantity management. Workflow automation is operationalized through SYS-1.3.4 (Automatic Process Flow), which ensures lifecycle progression without manual intervention. In addition to functional requirements, important non-functional and design-related requirements are considered, including configurability-related requirements (e.g., user-configurable routers and BOMs) and SYS-1.3.3 (Access Control), which introduces role-based security constraints. Data retention requirements further impose lifecycle considerations on storage architecture.

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#	Id	Name	Text
1		☐ ☒ System Requirements (SN-1)	
2	SNR-1.4	☐ ☒ SNR-1.4 Tracked Assets	Assets shall be tracked.
3	SYS-1.4.1	☐ ☒ SYS-1.4.1 Unique ID Assets	A unique ID shall be assigned to each asset.
4	SYS-1.4.1.1	☒ SYS-1.4.1.1 Searchable Assets	Assets shall be searchable by unique ID in the system.
5	SYS-1.4.1.2	☒ SYS-1.4.1.2 Field for Asset ID	Routers shall include a field for asset unique ID.
6	SNR-1.3	☐ ☒ SNR-1.3 Automated Workflows	Workflows shall be automated, but configurable to the project needs.
7	SYS-1.3.4	☐ ☒ SYS-1.3.4 Automatic Process Flow	System shall automatically flow through the process flow.
8	SYS-1.3.4.1	☒ SYS-1.3.4.1 Router Components ID	Only components for that router shall be issued to the unique ID.
9	SYS-1.3.4.2	☒ SYS-1.3.4.2 BOM Loading	BOM shall be loaded upon opening the router for that process step.
10	SYS-1.3.3	☐ ☒ SYS-1.3.3 Access Control	System shall have access level controls.
11	SYS-1.3.3.1	☒ SYS-1.3.3.1 BOM Upload	Owners shall be the only user to upload a BOM.
12	SYS-1.3.3.2	☒ SYS-1.3.3.2 Process Flow Upload	Owners shall be the only user to upload a process flow.
13	SYS-1.3.3.3	☒ SYS-1.3.3.3 Router Upload	Owners shall be the only user to upload router.
14	SYS-1.3.2	☐ ☒ SYS-1.3.2 Data Storage	System shall store data.
15	SYS-1.3.2.1	☒ SYS-1.3.2.1 Configurable Component	Component Quantity shall be configurable by user.
16	SYS-1.3.2.2	☒ SYS-1.3.2.2 Configurable Data Fields	Data fields shall be configurable.
17	SYS-1.3.2.3	☒ SYS-1.3.2.3 Data Storage Lifetime	Data shall be stored for the lifetime of the project or as defined by user.
18	SYS-1.3.1	☐ ☒ SYS-1.3.1 Created Workflows	System shall accept user created workflows.
19	SYS-1.3.1.1	☒ SYS-1.3.1.1 Configurable Routers	Routers shall be configurable by user.
20	SYS-1.3.1.2	☒ SYS-1.3.1.2 Configurable BOMS	BOM shall be configurable by user.
21	SNR-1.2	☐ ☒ SNR-1.2 Inventory Lookup	User shall be able to pull from current inventory.
22	SYS-1.2.1	☐ ☒ SYS-1.2.1 Searchable Inventory	System shall allow each component inventory and location to be searchable.
23	SYS-1.2.1.6	☒ SYS-1.2.1.6 Physical Storage Location	Each digital storage location shall have an associated physical storage location.
24	SYS-1.2.1.5	☒ SYS-1.2.1.5 Associated Projects	System shall find associated projects based on part number identification.
25	SYS-1.2.1.4	☒ SYS-1.2.1.4 Digital Storage Location Identification	System shall find component storage location based part number identification.
26	SYS-1.2.1.3	☒ SYS-1.2.1.3 Digital Storage Location	Components shall be stored in a digital storage location based on part number identification.
27	SYS-1.2.1.2	☒ SYS-1.2.1.2 Part Number Identification Association	Each part number identification shall be associated with at least one project.
28	SYS-1.2.1.1	☒ SYS-1.2.1.1 Part Number Identification	Components shall have a part number identification.
29	SNR-1.1	☐ ☒ SNR-1.1 Tracked Inventory	Inventory shall be tracked.
30	SYS-1.1.1	☐ ☒ SYS-1.1.1 Track Component Quality	System shall track component quantity.all be tracked.
31	SYS-1.1.1.5	☒ SYS-1.1.1.5 Stored Completed Assembly Quantity	Completed assembly quantity shall be stored for each unique ID.
32	SYS-1.1.1.4	☒ SYS-1.1.1.4 Configurable Component Quantity	Component quantity shall be configurable by the user.
33	SYS-1.1.1.3	☒ SYS-1.1.1.3 Stored Component Quantity	Component quantity shall be stored for each Unique ID.
34	SYS-1.1.1.2	☒ SYS-1.1.1.2 Inventory Sync	System shall decrease the amount of available inventory upon issuance to a build.
35	SYS-1.1.1.1	☒ SYS-1.1.1.1 Unique ID	A unique ID shall be assigned to each component or lots of components based on the project need.

Figure 2. Derived System Requirements Table for Stakeholder Need 1

Figure 3 presents the system requirements derived from Need 2's stakeholder requirements. Requirements such as SYS-2.2.1 (Component Origin Information) and SYS-2.4.1 (Component Data History) form the backbone of traceability by mandating storage of supplier, purchase order, manufacturing, and inspection data, while SYS-2.2.1.3 (Component Information Storage) and SYS-2.2.1.4 (Stored Data Entry) introduce non-functional audit and retention constraints that ensure data persistence and defensibility. Criticality designation requirements, including SYS-2.1.1 (User Designation) and visual status indicators, operationalize risk awareness by clearly flagging components requiring heightened oversight. Accessibility requirements such as SYS-2.3.1 (User Component Accessibility) ensure traceability is not merely stored but readily retrievable from Bills of Materials (BOM) or kit views, reinforcing usability expectations.

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#	Id	Name	Text
1		☐ ☒ R System Requirements (SN-2)	
2	SNR-2.1	☐ ☒ R SNR-2.1 Component Provenance	The system shall tie supplier, PO, and receiving inspections to component provenance or history.
3	SYS-2.1.1	☐ ☒ R SYS-2.1.1 User Designation	The system shall allow a user to designate a component as critical to a work flow or process.
4	SYS-2.1.1.2	☒ R SYS-2.1.1.2 Component Status	The system shall display a components critical status when that component is listed on a BOM or kit view.
5	SYS-2.1.1.1	☒ R SYS-2.1.1.1 Component Information Display	The system shall display a visual indicator on the component information page to alert users that the component has been labeled critical.
6	SNR-2.2	☐ ☒ R SNR-2.2 Component History Log	The system shall store component history.
7	SYS-2.2.1	☐ ☒ R SYS-2.2.1 Component Origin Information	The system shall store component history and origin information.
8	SYS-2.2.1.1	☒ R SYS-2.2.1.1 Supplier Provided Information	The system shall accept supplier or manufacturer provided information about a components origins and manufacturing history.
9	SYS-2.2.1.3	☒ R SYS-2.2.1.3 Component Information Storage	The system shall store all provided manufacturing and origin information associated with a component and archive that information if a component is removed from active use or deleted from the system.
10	SYS-2.2.1.2	☒ R SYS-2.2.1.2 User Provided Information	The system shall accept user or GTRI provided information about a components origins or manufacturing history.
11	SYS-2.2.1.4	☒ R SYS-2.2.1.4 Stored Data Entry	The system shall store all data entry events for traceability and to ensure data-retention.
12	SNR-2.3	☐ ☒ R SNR-2.3 Component History Accessibility from BOM	Component history shall be accessible from the BOM.
13	SYS-2.3.1	☐ ☒ R SYS-2.3.1 User Component Accessibility	The system shall allow for a user to access a component's history from the BOM or kit-build views.
14	SYS-2.3.1.2	☒ R SYS-2.3.1.2 Display Information	The system shall display a list of all BOMs, kits, and workflows a component is assigned to at a given time.
15	SYS-2.3.1.1	☒ R SYS-2.3.1.1 Component Navigation	The system shall allow a user to navigate to a component detail and history view from the BOM list or kit-build.
16	SNR-2.4	☐ ☒ R SNR-2.4 Critical Components Identification	The system shall identify critical components.
17	SYS-2.4.1	☐ ☒ R SYS-2.4.1 Component Data History	The system shall store information about a components purchase history, supplier data, associated POs, and receiving inspection results.
18	SYS-2.4.1.1	☒ R SYS-2.4.1.1 Inspection Results Workflow	The system shall allow for receiving inspection results to be added for each component to ensure full traceability.
19	SYS-2.4.1.2	☒ R SYS-2.4.1.2 Specific Component Data History	The system shall store specific component data sourced from PO and purchasing documents.

Figure 3. Derived System Requirements Table for Stakeholder Need 2

Derived system requirements traced from Need 3 (shown in Figure 4) help to operationalize controlled kit creation and configuration integrity through structured workflow and validation mechanisms. Core functional requirements such as SYS-3.1.1 (Kit Creation) and SYS-3.1.2 (BOM Validation) ensure that kits are instantiated only from approved and current Bills of Materials (BOMs), reinforcing configuration discipline. Requirements including SYS-3.2.1 (Inventory Availability Check) and SYS-3.2.2 (Inventory Reservation) translate stakeholder expectations into automated safeguards that prevent allocation conflicts and build delays. Downstream execution is supported by SYS-3.2.3 (Pick List Generation) and SYS-3.3.1 (Assembly Tracking), which provide structured assembly visibility, while SYS-3.3.3 (Configuration Integrity) and SYS-3.3.4 (Component Traceability) enforce alignment between kits, components, and approved BOM versions. Although most requirements are functional, SYS-3.4.1 (Role-Based Access Control) introduces a non-functional security constraint to protect configuration authority and maintain governance.

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#	Id	Name	Text
1		☐ ☒ R System Requirements (SN-3)	
2	SNR-3.4	☐ ☒ R SNR-3.4 Audit Logging	The system shall maintain logs of kit creation, modification, and issuance events.
3	SYS-3.4.1	☒ R SYS-3.4.1 Access Control	The system shall enforce role-based access control for all user actions.
4	SNR-3.3	☐ ☒ R SNR-3.3 Assembly Status Visibility	The system shall provide visibility into kit assembly status throughout preparation.
5	SYS-3.3.3	☒ R SYS-3.3.3 Configuration Integrity	The system shall detect and flag deviations from the approved BOM configuration.
6	SYS-3.3.4	☒ R SYS-3.3.4 Component Traceability	The system shall maintain traceability between kits, components, BOM versions, and associated projects.
7	SYS-3.3.2	☒ R SYS-3.3.2 Kit Issuance	The system shall allow authorized users to issue completed kits to designated build teams.
8	SYS-3.3.1	☒ R SYS-3.3.1 Assembly Tracking	The system shall track kit assembly status at the component level.
9	SNR-3.2	☐ ☒ R SNR-3.2 Component Availability	The system shall verify component availability prior to kit preparation.
10	SYS-3.2.2	☒ R SYS-3.2.2 Inventory Reservation	The system shall reserve components assigned to a kit to prevent allocation conflicts.
11	SYS-3.2.1	☒ R SYS-3.2.1 Inventory Availability Check	The system shall automatically check inventory availability for all required components before confirming kit creation.
12	SYS-3.2.3	☒ R SYS-3.2.3 Pick List Generation	The system shall generate a pick list identifying required components, quantities, and storage locations.
13	SNR-3.1	☐ ☒ R SNR-3.1 Kit Creation Capability	The system shall maintain logs of kit creation, modification, and issuance events.
14	SYS-3.1.2	☒ R SYS-3.1.2 BOM Validation	The system shall validate that the selected BOM is current and approved prior to kit creation.
15	SYS-3.1.1	☒ R SYS-3.1.1 Kit Creation	The system shall allow authorized users to create a kit instance from an approved BOM.

Figure 4. Derived System Requirements Table for Stakeholder Need 3

Figure 5 shows derived system requirements SNR/Need 4, supporting compliance and contract artifact support within PICKS. Foundational functional requirements such as SYS-4.1.1 (Part-to-CDRL Association Capability) and SYS-4.1.2 (Automatic CDRL Flagging at Receipt) ensure that contract-driven documentation requirements are systematically tied to inventory records at the point of entry. Traceability is further strengthened through SYS-4.3.1 (User Attribution Logging) and SYS-4.3.2 (Timestamped Compliance Logging), while SYS-4.3.3 (Tamper-Evident Audit Log Storage) introduces a design constraint that impacts data architecture and audit defensibility. Reporting and enforcement mechanisms are captured in SYS-4.4.1 (Project-Based Compliance Report Generation) and SYS-4.5.1 (Release Block for Incomplete Documentation), with performance requirements such as SYS-4.4.2 (Report Generation Performance) and SYS-4.5.3 (Compliance Check Response Time) introducing measurable system responsiveness expectations. Interface requirements, including SYS-4.6.1 (External CDRL Metadata Import via API), define integration boundaries and secure data exchange with external systems. Notably, recent sponsor feedback indicated that while comprehensive compliance artifact tracking is strategically desirable, some of these requirements may be considered “nice to have” in the near term and may need to be relaxed/loosened for a minimum viable product in the next development phase, given the current maturity of existing compliance infrastructure and enterprise systems.

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#	Id	Name	Text
1		System Requirements (SN-4)	
2	SNR-4.1	SNR-4.1 CDRL-Based Item Identification	The system shall identify inventory items that require contract-driven documentation based on project-level Contract Data Requirements Lists (CDRLs).
3	SYS-4.1.2	SYS-4.1.2 Automatic CDRL Flagging at Receipt	The system shall automatically flag inventory items associated with a CDRL when they are received into inventory.
4	SYS-4.1.1	SYS-4.1.1 Part-to-CDRL Association Capability	The system shall allow authorized users to associate specific part numbers with one or more CDRL identifiers.
5	SNR-4.2	SNR-4.2 Receiving Record to Deliverable Linkage	The system shall link receiving and inspection records to applicable contract deliverable artifacts.
6	SYS-4.2.2	SYS-4.2.2 Exportable Compliance Artifact Records	The system shall export linked receiving and artifact records in a structured digital format (e.g., CSV or PDF).
7	SYS-4.2.1	SYS-4.2.1 Receiving-to-CDRL Artifact Linking	The system shall allow users to associate a receiving record with a specific CDRL artifact record.
8	SNR-4.3	SNR-4.3 Compliance Action Auditability	The system shall maintain an auditable record of compliance-related actions performed on inventory items.
9	SYS-4.3.3	SYS-4.3.3 Tamper-Evident Audit Log Storage	The system shall store compliance audit logs in a tamper-evident format.
10	SYS-4.3.2	SYS-4.3.2 Timestamped Compliance Logging	The system shall record a timestamp for each compliance-related action.
11	SYS-4.3.1	SYS-4.3.1 User Attribution Logging	The system shall record the user ID for each compliance-related action.
12	SNR-4.4	SNR-4.4 Contract-Compliant Report Generation	The system shall generate contract-compliant compliance reports suitable for sponsor delivery or audit readiness.
13	SYS-4.4.1	SYS-4.4.1 Project-Based Compliance Report Generation	The system shall allow authorized users to select a project and generate a compliance report.
14	SYS-4.4.2	SYS-4.4.2 Compliance Report Generation Performance	The system shall generate a compliance report for a project containing up to 5,000 inventory records within 30 seconds.
15	SYS-4.4.3	SYS-4.4.3 CDRL Inclusion in Reports	The system shall include all linked CDRL identifiers in the generated compliance report.
16	SNR-4.5	SNR-4.5 Compliance-Gated Inventory Release	The system shall enforce completion of required compliance documentation prior to inventory release for use.
17	SYS-4.5.1	SYS-4.5.1 Release Block for Incomplete Documentation	The system shall prevent status transition of a flagged item to "Released" if required documentation is incomplete.
18	SYS-4.5.2	SYS-4.5.2 Release Block Notification	The system shall display a notification when release is blocked due to missing compliance documentation.
19	SYS-4.5.3	SYS-4.5.3 Compliance Check Response Time	The system shall evaluate compliance completeness prior to release within 1 second of user action.
20	SNR-4.6	SNR-4.6 External Compliance Metadata Integration	The system shall support integration of contract compliance metadata from external enterprise systems.
21	SYS-4.6.1	SYS-4.6.1 External CDRL Metadata Import via API	The system shall import CDRL identifiers and associated metadata from an external system via secure application programming interface (API).
22	SYS-4.6.2	SYS-4.6.2 Compliance Metadata Schema Validation	The system shall validate imported compliance metadata against defined schema rules prior to storage.

Figure 5. Derived System Requirements Table for Stakeholder Need 4

Derived system requirements stemming from Need 5 (shown in Figure 6) define the integration expectations between PICKS and Workday and emphasize non-functional qualities needed for dependable enterprise interoperability. Core functional requirements such as SYS-5.1.1 (Initiate Workday Lookup) and SYS-5.3.1 (Auto-Populate Receiving Fields) reduce manual entry and improve data accuracy, while SYS-5.3.1.1 (Documented Field Mapping) introduces an integration design requirement that supports maintainability and consistent data semantics across systems. This need is heavily driven by non-functional requirements: performance constraints (e.g., SYS-5.2.1 specifying retrieval within 5 seconds for 95% of valid requests and timely display for confirmation), reliability requirements such as SYS-5.5.1 requiring user notification and a manual fallback workflow when Workday is unavailable, and security constraints including SYS-5.6.1 mandating Transport Layer Security (TLS) 1.2 or higher for external application programming interface (API) communications. Finally, audit/compliance requirements such as SYS-5.7.1 require complete logging of Workday lookup transactions and storage/export of interface transaction records, which ensures defensible traceability for enterprise operations.

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#	Id	Name	Text
1		☐ R System Requirements (SN-5)	
2	SNR-5.1	☐ R SNR-5.1 Workday Lookup	The System shall accept a PO number from the user and initiates a Workday lookup.
3	SYS-5.1.1	☐ R SYS-5.1.1 User Initiation Lookup	PICKS shall accept a user-entered PO identifier and initiate a Workday lookup within 1 second of submission.
4	SYS-5.1.1.1	☐ R SYS-5.1.1.1 Lookup Interface	User interface shall provide a PO entry field with input validation and lookup trigger.
5	SYS-5.2	☐ R SYS-5.2 Retrieve Workday Data	The System shall retrieve PO data from Workday within an acceptable response time.
6	SYS-5.2.1	☐ R SYS-5.2.1 Data Retrieval	PICKS shall retrieve vendor name, vendor ID, line items, and quantities from Workday within 5 seconds for 95% of valid PO lookup requests.
7	SYS-5.2.1.1	☐ R SYS-5.2.1.1 Workday Interface	The Workday interface shall support API response times sufficient to meet the 5-second retrieval requirement.
8	SNR-5.3	☐ R SNR-5.3 Populate Record Fields	The System shall use retrieved Workday data to populate receiving record fields.
9	SYS-5.3.1	☐ R SYS-5.3.1 Autopopulated Fields	PICKS shall auto-populate receiving record fields using Workday PO data, including vendor, line items, and quantities, when valid PO data is available.
10	SYS-5.3.1.1	☐ R SYS-5.3.1.1 Field Mapping	PICKS shall maintain a documented field mapping between Workday PO fields and receiving record fields.
11	SNR-5.4	☐ R SNR-5.4 Display Retrieved Data	The System shall display retrieved PO data to the user for confirmation before record creation.
12	SYS-5.4.1	☐ R SYS-5.4.1 Display Data for User	PICKS shall display retrieved PO data for user confirmation within 2 seconds of successful data retrieval.
13	SYS-5.4.1.1	☐ R SYS-5.4.1.1 Receiving Interface	Receiving interface shall include a confirmation view showing vendor and line items.
14	SNR-5.5	☐ R SNR-5.5 Fallback Workflow	The System shall provide a fallback workflow when the Workday interface is unavailable.
15	SYS-5.5.1	☐ R SYS-5.5.1 Manual Entry	If Workday does not respond within 10 seconds, PICKS shall notify the user and enable manual receiving entry within 2 seconds.
16	SYS-5.5.1.1	☐ R SYS-5.5.1.1 Manual Receiving Workflow	System shall provide a manual receiving workflow flagged as "Workday Not Verified."
17	SNR-5.6	☐ R SNR-5.6 Data Protection	The System shall protect data exchanged across the Workday interface.
18	SYS-5.6.2	☐ R SYS-5.6.2 Transaction Log	PICKS shall log all Workday PO lookup transactions with 100% audit capture, including user ID, PO number, timestamp, and outcome.
19	SYS-5.6.1	☐ R SYS-5.6.1 Encryption	PICKS shall encrypt all Workday interface communications using TLS 1.2 or higher.
20	SYS-5.6.2.1	☐ R SYS-5.6.2.1 Audit Subsystem	Audit subsystem shall store and export interface transaction records for authorized users.
21	SYS-5.6.1.1	☐ R SYS-5.6.1.1 TLS Encryption	Interface configuration shall enforce TLS encryption for all external API connections.

Figure 6. Derived System Requirements Table for Stakeholder Need 5

The derived requirements for Stakeholder Need 6 were created to refine the high-level enterprise integration objective into specific, testable capabilities while maintaining flexibility for future system decisions. Since the State of Georgia's ERP solution has not yet been selected or implemented, the team intentionally developed vendor-agnostic requirements that focus on the functions PICKS must support rather than a specific technology. The first requirement, SYS-6.1 ERP Interface, establishes that PICKS shall interface with the future ERP to retrieve inventory, asset, and procurement information. Derived requirements for this capability include supporting secure API-based communication, retrieving purchase order information, accessing inventory availability data prior to kit creation, retrieving asset status and ownership information, and notifying users when ERP data is unavailable. The second requirement, SYS-6.2 Inventory Synchronization, ensures that PICKS maintains up-to-date inventory and asset information through configurable synchronization intervals, discrepancy detection, synchronization logging, timestamping of synchronized records, and preservation of the most recent valid data during outages. The third requirement, SYS-6.3 Workflow Continuity, was developed to ensure that workflows remain operational even when the ERP is temporarily unavailable. Derived requirements include allowing users to continue workflows using cached ERP data, indicating when data is cached rather than live, queuing transactions for later synchronization, reconciling queued transactions after reconnection, and maintaining an audit trail of offline activities. Finally, SYS-6.4 Secure Data Exchange addresses cybersecurity and access concerns by requiring encrypted communication, authentication of users and services, role-based access control for ERP-derived information, audit

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logging of interface transactions, and prevention of unauthorized modification of ERP data within PICKS. Together, these derived requirements ensure that PICKS can leverage enterprise data without duplicating ERP functionality, allowing the system to focus on workflow automation, kitting, configuration management, and traceability while remaining scalable, resilient, and adaptable to future enterprise systems such as Oracle, Workday, or PEDYN.

Id	△ Name	Text
SNR-6	R SNR-6 Enterprise Integration	
6.1	R 6.1 Enterprise System	The system shall exchange data with enterprise systems to support workflow automation, traceability, and inventory visibility.
6.1.1	R 6.1.1 Retrieve Inventory from E	The system shall retrieve inventory availability data from the ERP prior to kit creation.
6.1.2	R 6.1.2 Retrieve Asset Status	The system shall retrieve asset status and ownership information from the ERP.
6.1.3	R 6.1.3 Notification if ERP Error	The system shall notify users if ERP data cannot be retrieved.
6.2	R 6.2 Inventory Synchronization	The system shall synchronize inventory availability and asset status with approved enterprise systems.
6.2.1	R 6.2.1 Synchronize Inventory	The system shall synchronize inventory quantities at a configurable interval.
6.2.2	R 6.2.2 Inventory Records	The system shall identify discrepancies between local and ERP inventory records.
6.2.3	R 6.2.3 Synchronization Failures	The system shall log synchronization failures and notify administrators.
6.2.4	R 6.2.4 Synchronization Timesta	The system shall timestamp synchronized inventory records.
6.3	R 6.3 Enterprise Resource Planning I	The system shall interface with a Enterprise Resource Planning (ERP) system to retrieve inventory, asset, and procurement information.
6.3.1	R 6.3.1 Cached ERP Data	The system shall allow users to continue workflow activities using cached ERP data.
6.3.2	R 6.3.2 Records Synchronization	The system shall queue ERP transactions for synchronization after connectivity is restored.
6.3.3	R 6.3.3 Reconnection of records	The system shall reconcile queued transactions with ERP records after reconnection.
6.3.4	R 6.3.4 Audit Log	The system shall record all offline workflow actions in an audit log.
6.4	R 6.4 Workflow Continuity	The system shall continue workflow execution when enterprise systems are temporarily unavailable.
6.4.1	R 6.4.1 Prevent Unauhtorized Mc	The system shall prevent unauthorized modification of ERP data within PICKS.
6.4.2	R 6.4.2 Record ERP Transactions	The system shall record all ERP interface transactions in an audit log.
6.4.3	R 6.4.3 Authenticate Users	The system shall authenticate ERP users and services before exchanging information.
6.4.4	R 6.4.4 Encrypt Data	The system shall encrypt all data exchanged with the ERP.

Figure 7. Derived System Requirements Table for Stakeholder Need 6

In the next phase of the project, the team will continue to review and refine these system requirements with the GTRI sponsors as additional clarification and enterprise context become available. Any new information regarding operational constraints, compliance maturity, or integration boundaries will be incorporated to further mature and prioritize the requirements set. At this stage, these defined stakeholder and system requirements have provided a structured foundation for exploring initial architectural options for PICKS and have directly informed early behavioral and functional analysis.

4.4 Requirements Verification

For all PICKS requirements, the team defined appropriate verification approaches and documented them within MSOSA. This process ensured that each requirement is both testable and verifiable. The approaches, organized by requirement set for each stakeholder need, are detailed below.

4.4.1 Inventory Visibility and Workflow Support Requirements Verification Approach

The verification associated with inventory visibility and workflow support is broken down into 4 main tests for the stakeholder needs associated. Within each test, there are multiple pass/fail determinations that determine the disposition of the system. The verification for component tracking starts with loading a simulated BOM and associating that BOM to a project. First gate checks that each component

has a unique part number associated with that component and the project number has been associated with the BOM. A simulation utilizing a batch of components being received will be performed. The quantity, unique number, and storage location for the component will be verified. A second lot of components will be simulated through the system to confirm that a different unique number is assigned. A de-bonus activity where the quantity is decreased is simulated to demonstrate that the user can increase and decrease the quantity. This concludes the incoming testing.

A manufacturing simulation shall be used to test the automated workflow capabilities. A BOM and router will be uploaded into the system. The manufacturing process will be simulated by utilizing two unique builds. The first build will be simulated through completion. The component for each operation will be verified as correct per the BOM. The components are issued to the unique ID for the system. The component unique IDs will be verified to determine the correct unique ID was added to the system or assembly records for each component. Reconciliation between the component inventory and quantity issued to the build will be verified. The second build will pause at an intermediate station. It will not go through completion. This is to simulate an incomplete system build. There are several verifications to occur during the simulation to test the PICKS system that meets the requirements associated with manufacturing.

The manufacturing simulation and incoming simulation shall be used to test the search capabilities. Utilizing the uploaded BOMs, components, simulated manufacturing and incoming, the following will be searched:

- Quantity for a unique ID
- Storage location(s) for a part number
- Storage location(s) for a unique ID
- Associated project(s) for a component
- Storage location(s)
- Completed systems quantity
- “Where used” for components based on unique ID

The information gathered from each search is verified against the expected results.

The last simulation is for user access control. In this simulation, two groups of users will be created associated with different projects. Each group should have access to only the assigned project. Both group will try to upload and change information for both projects. If the system allows only the group associated with that project to update information, but does not permit the non-associated group to upload information, then the user access control is verified.

Two requirements were verified via analysis or demonstration. The data storage requirement (SYS 1.3.2.3) will be set based on either the IT requirement or storage capacity. Analysis will show the storage limit meets the requirement. The digital storage location association with a physical location (SYS 1.2.1.6) will be performed by via demonstration. Digital storage locations will be setup based on the physical storage locations. This association and setup will be shown to the customer.

4.4.2 Component Traceability Requirements Verification Approach

The requirements for component traceability as derived from Stakeholder Need 2 will be verified through traceability-driven requirements. This begins with confirming that the system can correctly

capture, store, and retrieve the component-level data mandated by SYS-2.2.1 (Component Origin Information) and SYS-2.4.1 (Component Data History). Verification activities will include database inspection, API interrogation, and workflow walkthroughs to ensure supplier details, purchase order references, manufacturing data, and inspection records are captured at the moment of receipt or data entry. Test cases will validate that each required data field is present, correctly formatted, and linked to the appropriate unique component identifier. These verification tests collectively demonstrate that the foundational traceability data model behaves as specified and supports end-to-end genealogy.

Non-functional requirements such as SYS-2.2.1.3 (Component Information Storage) and SYS-2.2.1.4 (Stored Data Entry) require verification methods that focus on persistence, auditability, and data integrity. This includes testing retention rules, validating that audit logs are generated for all create/update/delete actions, and confirming that historical records cannot be overwritten or deleted without proper authorization. Verification will also include simulated failure scenarios, such as system restarts or interrupted processes, to ensure that data remains intact and auditable. These activities confirm that the system not only stores the required information but does so in a manner consistent with compliance and governance expectations.

Verification of accessibility and criticality-related requirements, including SYS-2.1.1 (User Designation) and SYS-2.3.1 (User Component Accessibility), will rely on user-focused demonstration and testing. This includes operational walkthroughs to ensure that users can designate criticality levels, view visual status indicators, and retrieve component information directly from Bills of Materials (BOMs), kit views, or search interfaces. Testers will validate that criticality flags appear consistently across all relevant screens and that traceability data is accessible within the expected number of clicks or navigation steps. These verification activities ensure that the system not only stores traceability information but presents it in a usable, intuitive manner that supports real-time decision-making and risk awareness.

4.4.3 Kitting and Configuration Management Requirements Verification Approach

To ensure the system meets Stakeholder Need 3—effective kitting and configuration management, each requirement in this section is verified using targeted methods that validate the full kit lifecycle from creation through issuance. For SYS-3.1.1 (Kit Creation) and SYS-3.1.2 (BOM Validation), verification is performed through demonstration and testing by having an authorized user attempt to create a kit from both approved and unapproved BOMs; the system must successfully create kits only from valid configurations and reject outdated ones. SYS-3.2.1 (Inventory Availability Check) and SYS-3.2.2 (Inventory Reservation) are verified through test scenarios where multiple kits request the same limited components—ensuring the system correctly identifies shortages and prevents overallocation by reserving inventory for the first valid request. For SYS-3.2.3 (Pick List Generation), demonstration confirms that the system generates an accurate pick list including quantities and storage locations immediately after kit creation. SYS-3.3.1 (Assembly Tracking) and SYS-3.3.2 (Kit Issuance) are verified through workflow demonstrations showing real-time status updates at the component level and restricting kit issuance to authorized users only. SYS-3.3.3 (Configuration Integrity) is tested by attempting to introduce incorrect or substituted components, where the system must flag and prevent deviations from the approved BOM. SYS-3.3.4 (Component Traceability) is verified through analysis and demonstration by retrieving a complete history of a component—from procurement to kit assignment to final build—within an acceptable timeframe. Finally, SYS-3.4.1 (Access Control) is validated through

demonstration testing that confirms role-based permissions are enforced and unauthorized actions are denied and logged. Together, these verification methods were chosen to ensure the system not only functions correctly but also achieves the stakeholder objective of providing accurate, controlled, and fully traceable kitting processes.

4.4.4 Compliance Artifacts Requirements Verification Approach

For Need 4, which focuses on contract-compliant report generation, the team selected a combination of demonstration and test-based verification methods to ensure both functional correctness and performance compliance of the system. Demonstration was used where user interaction and system behavior could be directly observed, such as in SYS-4.4.1 (Project-Based Compliance Report Generation). In this case, verification involves walking through the workflow of selecting a project and generating a compliance report, confirming that the system produces the expected output for a representative use case. This approach ensures that the system meets stakeholder expectations for usability and correct execution of core functionality.

For more data-intensive and measurable requirements, test-based methods were applied to validate system outputs and performance under controlled conditions. For example, SYS-4.4.2 (Compliance Report Generation Performance) is verified by populating the system with a test dataset (e.g., up to 5,000 inventory records) and measuring whether the report is generated within the required 30-second threshold. Similarly, SYS-4.4.3 (CDRL Inclusion in Reports) uses testing to compare the generated report content against the underlying database records to ensure all required CDRL identifiers are accurately included. These methods provide objective evidence that the system meets both functional accuracy and performance requirements.

4.4.5 Interface Definitions/Readiness Requirements Verification Approach

The Interface Definitions/Readiness requirements are verified using a mix of testing, inspection, and demonstration depending on what the requirement is trying to prove. The focus is on making sure the system correctly connects to and exchanges data with external systems like Workday. These verification methods apply across requirements such as SYS-5.1.1 (User Initiation Lookup), SYS-5.2.1 (Data Retrieval), SYS-5.3.1 (Auto-populated Fields), and SYS-5.5.1 (Manual Entry), which define the core interface behavior.

Most of the interface behavior, such as sending requests, retrieving data, and populating fields, is verified through testing. This means running the system and confirming that it interacts with the external system the way it is supposed to. For example, when a user performs a lookup, the system should return the correct data and populate the appropriate fields. Testing is also used to confirm that the system handles failures properly, such as when the external system is unavailable and a fallback process is required.

Some interface requirements are verified through inspection instead of execution. This includes checking field mappings, data formats, and interface definitions. This approach supports requirements such as SYS-5.3.2 (Field Mapping) and SYS-5.4.2 (Receiving Interface), where correctness is based on alignment between systems rather than runtime behavior. In these cases, the goal is to confirm that the internal system is aligned with the external system and that the data being exchanged is structured correctly. This is done by reviewing configurations and interface specifications rather than running the system.

User-facing interactions that depend on external data, such as displaying retrieved information or supporting manual entry during failures, can also be verified through demonstration. This aligns with requirements such as SYS-5.4.1 (Display Data for User) and SYS-5.5.2 (Manual Receiving Workflow), where the system behavior is best validated from the user's perspective. This helps confirm that the system behaves correctly and that the interface is usable and clear.

Security requirements tied to external interfaces, such as encryption, secure data transfer, and logging, are primarily verified through inspection. This includes requirements such as SYS-5.6.1 (Encryption), SYS-5.6.3 (TLS Encryption), and SYS-5.6.4 (Audit Subsystem), ensuring that the correct controls are in place and properly configured. In some cases, logs or system outputs may be reviewed to confirm that these controls are working as expected.

This approach keeps verification focused and practical. It ensures that external interfaces function correctly, data is exchanged accurately, and integrations remain secure, while using the most appropriate method for each type of requirement.

4.5 Requirements Prioritization

All PICKS requirements were prioritized using a weighted scoring approach to identify which capabilities are most important for the MVP and longer-term architecture. Each requirement was scored across three dimensions: mission alignment, MVP operational criticality, and complexity penalty. Mission alignment was weighted highest because requirements that directly support enterprise-wide visibility, traceability, compliance, and standardized workflows are central to the value of PICKS. MVP operational criticality was also weighted strongly to highlight requirements that are necessary for core workflows such as receiving, inventory tracking, kitting, and lifecycle execution.

The complexity penalty was included to keep the prioritization realistic by accounting for implementation difficulty, integration burden, and development effort. This allowed requirements that are highly valuable but overly complex to be balanced against simpler requirements that can provide near-term operational impact. The final weighted scores were then mapped to priority tiers of Very High, High, and Medium, helping the team distinguish between requirements that are essential for an MVP, important for future increments, and useful but lower-priority capabilities. Figure showcases this prioritization approach in more detail.

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Dimension	Scale	Weight in Score	Interpretation
Mission Alignment	1 to 3	x4	Degree to which the requirement supports the strategic objectives of PICKS, such as enabling enterprise-wide visibility, traceability, compliance, and standardized workflows. This reflects how strongly the requirement contributes to delivering sponsor-defined value.
MVP Operational Criticality	1 to 3	x3	Degree to which the requirement is functionally necessary for core Minimum Viable Product (MVP) workflows to operate, such as receiving, inventory tracking, kitting, and lifecycle execution. Requirements that, if absent, would prevent or significantly disrupt these workflows are rated highest.
Complexity Penalty	1 to 3	-1x	Relative implementation difficulty, including technical complexity, integration burden, and development effort. This is applied as a moderated penalty to ensure that mission- and operationally-critical capabilities remain prioritized, even if they are complex to implement.

Scoring Formula	Priority Tier Thresholds
Weighted Score = (Mission Alignment x 4) + (MVP Operational Criticality x 3) - Complexity Penalty	Very High: 19-20
	High: 16-18
	Medium: 11-15

Figure 8. Requirements prioritization scoring framework for PICKS, showing the weighted dimensions, scoring scale, formula, and priority tier thresholds used to rank all requirements

For Need 1: Inventory Visibility and Workflow Support, the prioritization shows a strong concentration of requirements in the Very High tier, confirming that this need is foundational to the overall PICKS system. The highest-priority requirements emphasize data structure, inventory synchronization, location tracking, associated projects, workflows, and access control, showing that visibility depends on more than just search or UI features. The results suggest that PICKS should first establish standardized, opinionated workflows and reliable inventory data structures before adding broader flexibility or configurability later.

For Need 2: Component Traceability, the prioritization results show the strongest concentration of Very High priorities across all needs, indicating that traceability is one of the most critical capabilities for PICKS. The highest-ranked requirements emphasize persistent component history, origin/manufacturing data, stored data entries, inspection results, and lifecycle visibility, suggesting that PICKS needs a durable data backbone that links component records across events and workflows. Usability features such as tagging, access, and navigation are still important, but ranked lower than requirements that protect data integrity and ensure reliable traceability across the component lifecycle.

For Need 3: Kitting and Configuration Management, the prioritization results show that the highest-value requirements are centered on preventing incorrect builds through BOM validation, kit creation, inventory reservation, pick list generation, assembly tracking, and configuration integrity. This indicates

that PICKS must tightly connect inventory state with kitting logic so users can confirm availability, reserve the correct components, and maintain traceability between kits, BOM versions, and associated projects. Decision-support features remain important, while governance-related items like access control rank slightly lower because they are less directly tied to day-to-day kitting execution.

For Need 4: Compliance Artifacts, the prioritization shows a clear split between high-priority enforcement capabilities and lower-priority monitoring or integration features. Requirements tied to release blocking, early CDRL/documentation flagging, artifact linking, and compliance logging ranked highest because they embed compliance directly into the workflow instead of treating it as a separate after-the-fact activity. More complex capabilities, such as external metadata imports, report generation performance, and schema validation, ranked lower, suggesting they may be better suited for later MVP increments once the core compliance enforcement and audit foundation is in place.

For Need 5: Interface Definitions and Readiness, the prioritization shows a strong contrast between security-related requirements and broader performance or integration capabilities. Encryption, TLS enforcement, transaction logging, and audit support ranked highest because secure interface behavior is essential for compliance and cannot be traded off easily. In contrast, ERP lookup speed, field mapping, data retrieval, and receiving interface features generally ranked in the Medium tier, suggesting that slower or more manual integration workflows may be acceptable for the MVP as long as the system remains secure, auditable, and functionally usable.

4.6 High-Level Operational Concept View (OV-1)

A high-level OV-1 (Figure 9) was developed alongside the requirements which depicts PICKS as a centralized enterprise software capability that supports inventory visibility, kitting, traceability, and compliance across GTRI. While PICKS operates as a single system of record, it is shown through zone-specific segments or consoles to reflect how different user communities interact with it in distinct operational contexts. In the Receiving & Warehouse zone, procurement and logistics personnel use the receiving segment to record receipt, inspection, stocking, transfers, and order updates; in Labs & Engineering, engineers and technicians search, reserve, issue, and assemble kits, with Quality Assurance performing validation and release steps; and in Program/Enterprise Oversight, leadership accesses project readiness, on-order visibility, and key performance indicators. The Governance & Security segment supports system configuration, compliance artifact management, deployment and patching, and cybersecurity framework updates such as Cybersecurity Maturity Model Certification (CMMC) and National Institute of Standards and Technology (NIST) requirements. Externally, PICKS securely interfaces with Workday via APIs for purchase order and vendor reference data and generates audit evidence for government sponsors, while also enabling transition from legacy lab tools.

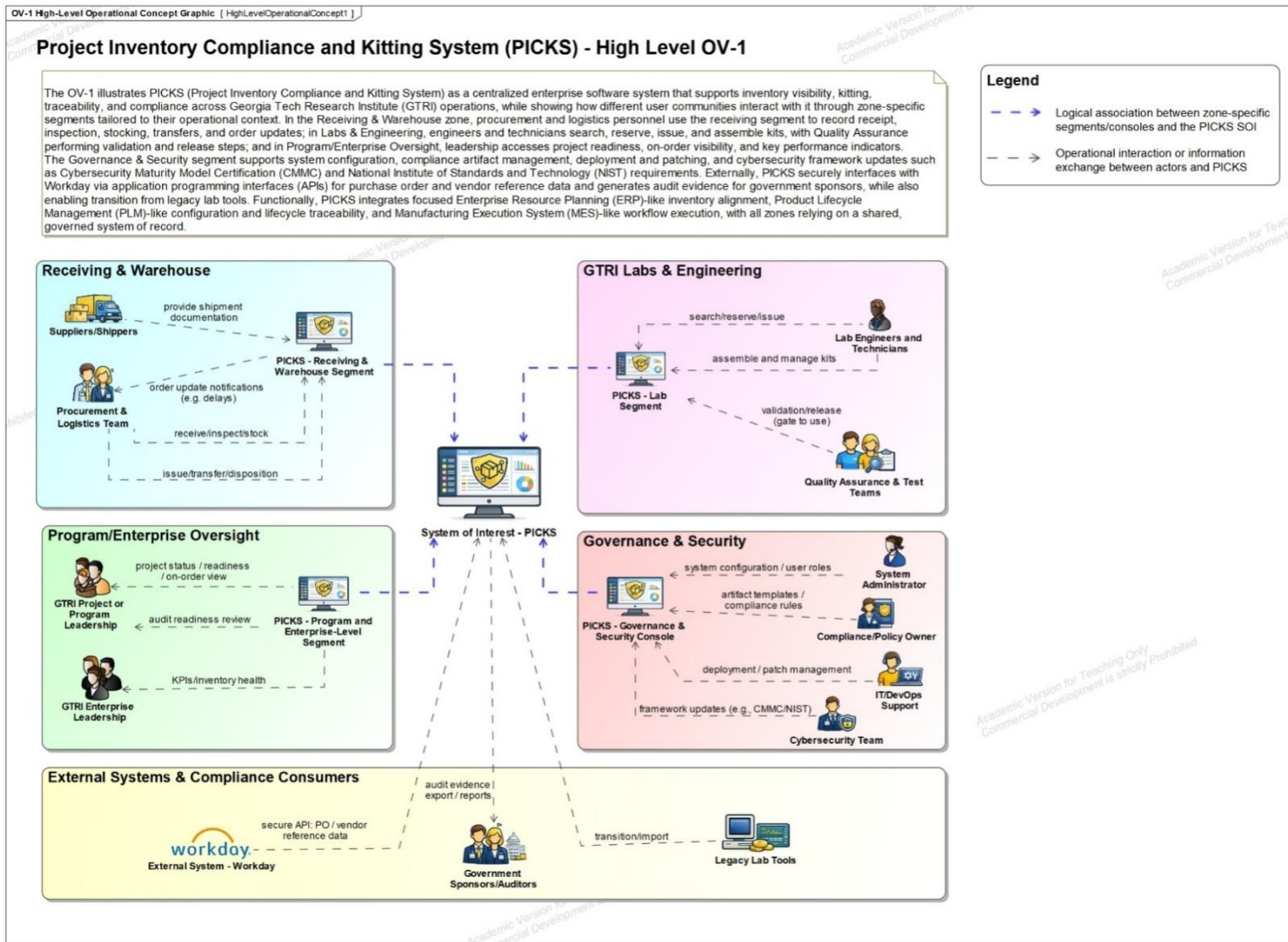


Figure 9. High-Level OV-1 of PICKS

5. Solution Development & Alternatives*

5.1. Discussion

Figure 10 shows the exchange of information from the typical Off-the-shelf (OTS) systems that are available in the marketplace. The ERP performs resource management including inventory management, generating work orders, cost, scheduling, etc. A MES system is the interphase with the operations personnel during production. This is where lab engineers and technicians add manufacturing data, perform material issuance, etc. A PLM manages the configurations, stores the specifications, drawings, models, etc. (ERP vs PLM vs MES: What's the Difference and Which One Does Your Industry Need?, 2025) Not all functionality from each core system is required to complete the objectives for the overall system. Options for the systems includes: Off the shelf (OTS), open source, full custom, or hybrid.

A high-level functional decomposition for PICKS is displayed in Figure 11 PICKS functional breakout shows the key functions. This functional decomposition was used to assist in the creation of the use cases. As part of this solution, additional functionality such as unique ID generation, part number generation, BOM framework, and router framework are required to be generated or leveraged from existing GTRI infrastructure.

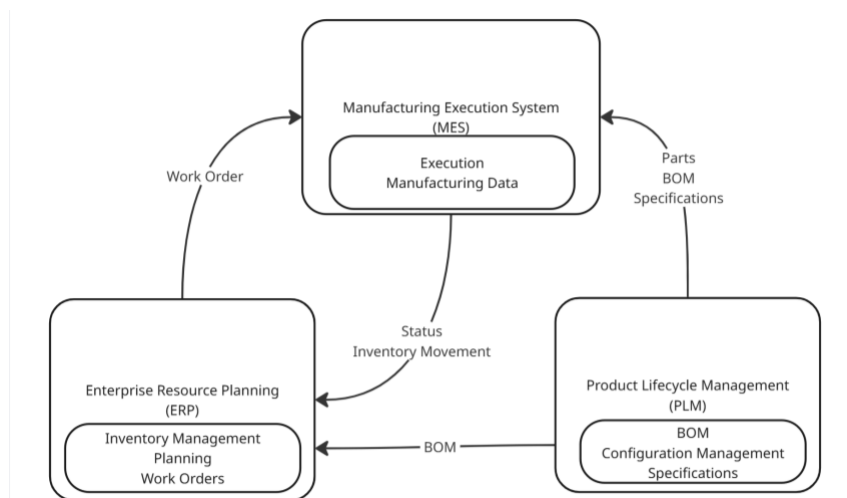


Figure 10 Core systems and functionality of interest for the PICKS project.

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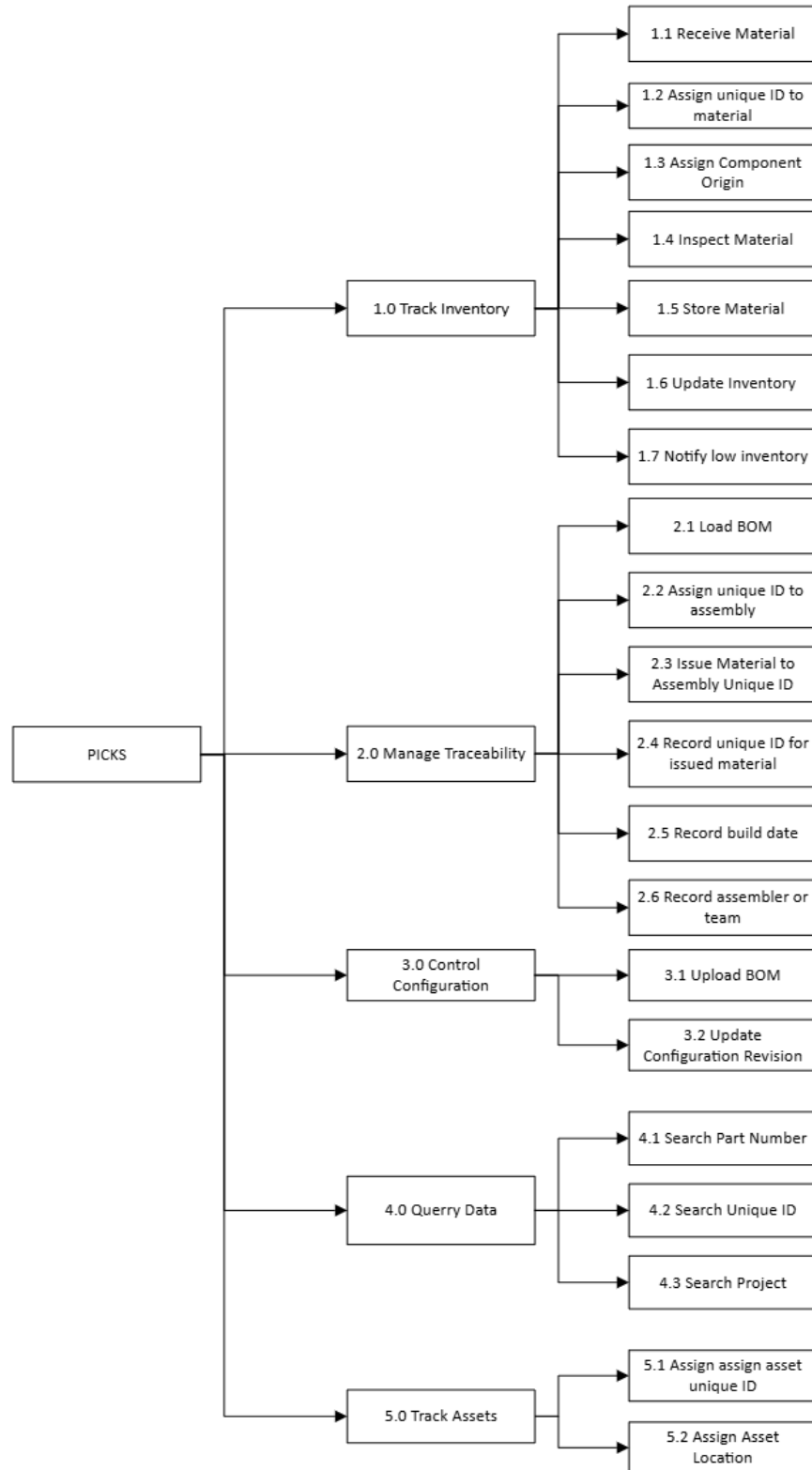


Figure 11 PICKS functional breakout

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As part of the solutions development phase, an evaluation of the existing core systems was performed via interviews with several labs. This included development of material flows based on site, identification of actors, and core systems each lab uses, see Figure 7 and Figure 12. Depending on the campus, different personnel handle the material, however all procurement occurs within Georgia Procurement Portal (GPP). This is where the purchase order is generated and financials are handled. Once received, any assets greater than \$5,000 will be flagged to be tagged and tracked via Sunflower. Both GPP and Sunflower are required systems set by either the state of Georgia or the university. All GTRI labs are required to use those systems. The other systems, found within the black boxes of Figure 7 and Figure 8, had been either created or customized of each lab. The driver for the use of additional systems came from project requirements and complexity.

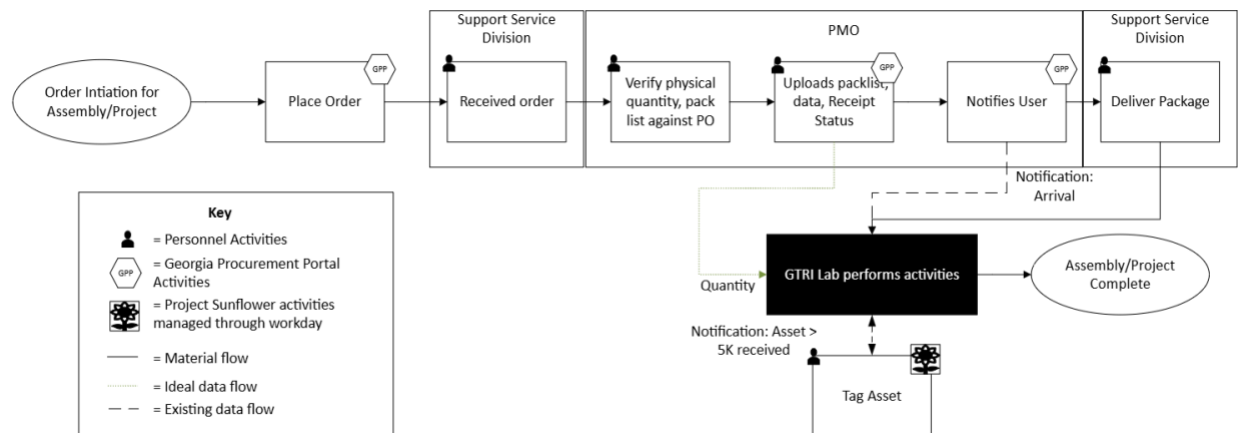


Figure 7: CCRF Campus Material Flow

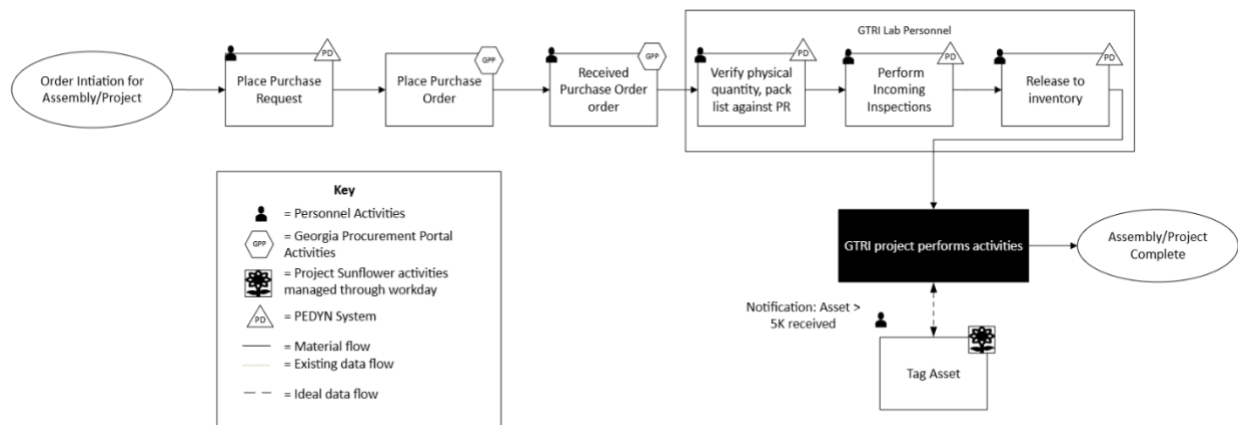


Figure 8 Headquarter Campus Material Flow

During conversations with the various labs, it was found that 3 labs utilize the same OTS software, PEDYN, as their ERP. One lab utilizes the OTS version while the other two utilize a hybrid custom version of PEDYN. The flow of data and materials is shown in Figure 9. Several challenges were identified with PEDYN including, the software is run through an archaic backend, everything is manual (no automated work flows, etc), and anyone with a license can edit the data. Hybrid version did automate the upload of the BOM into the PEDYN, however still had challenges with configuration management, workflow automation, and roles based access control (RBAC). To address RBAC concerns, some labs limit the

access to certain users to minimize the risk of unauthorized changes. Some parts of the automation that were implemented still have challenges and need additional verification to confirm the BOM was uploaded appropriately to PEDYN. The cumbersomeness of the system and challenges in training make it difficult to implement as is across the GTRI labs.

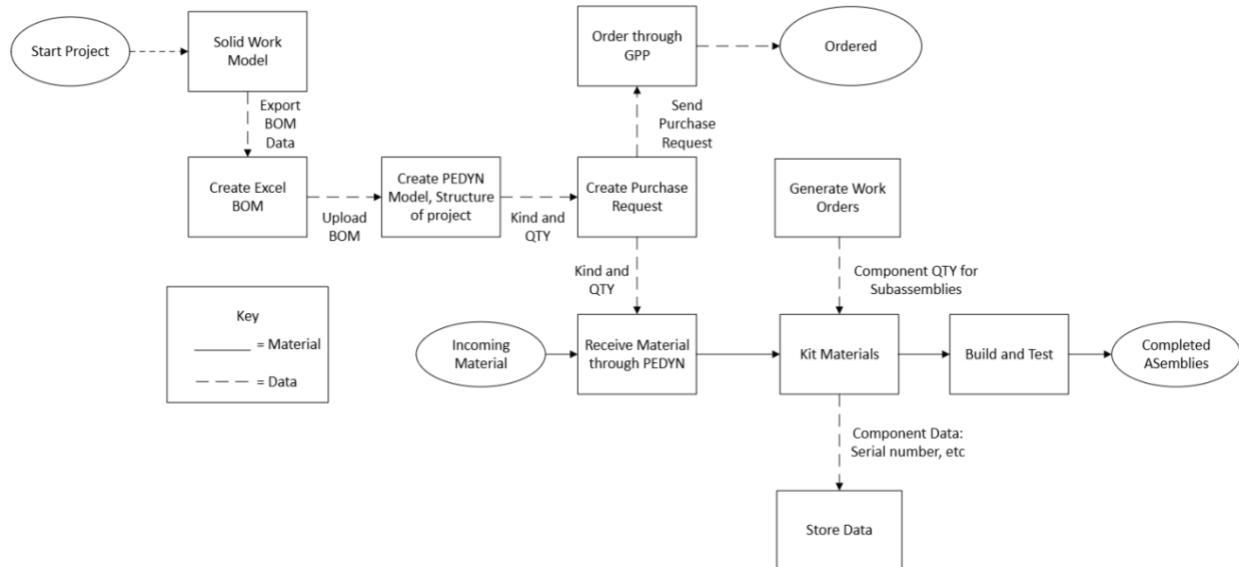


Figure 9 Break down of black box for lab utilizing PEDYN

During the solution generation phase, it was announced that state of Georgia would be implementing an ERP. This ERP would most likely use Oracle and/or workday to do both financial and inventory management of materials and assets. The implementation of the system will be after the capstone completion. The assumption the team and mentors are proceeding with is that the ERP system will meet the minimum (standard) requirements of an ERP. This ERP will be factored into the final design of PICKS. The team solutions will prioritize the other high priority functionalities such as automated workflows and RBAC protocols. A menu of open source and OTS MES systems will be proposed with integration information for future integration to the future ERP or PEDYN. This allows for the GTRI to determine which option will best suited for their needs when all information is available. At this time, the team is not focusing on PLM functionality as this may be met with the other core systems (ERP or MES).

5.2. Use Case

The final system (integration of the MES and ERP systems with RBAC) architecture was developed as part of this capstone. These use cases apply to the final integrated system, not the MES standalone system.

The inventory tracking use case, Figure 10 Inventory Tracking Use Case, focuses on inventory management. When the material is received, a unique ID is assigned. Additional data based on project requirements and system requirements are added by receiving personnel. This data is stored in the system. The user updates the received quantity in the system. Component quantity is increased or decreased based on addition or removal from inventory. Project personnel can review the status and inventory for each component. The relevant displayed information includes the part number, unique ID,

physical location, and associated projects. When the component is low, the system will notify the purchaser.

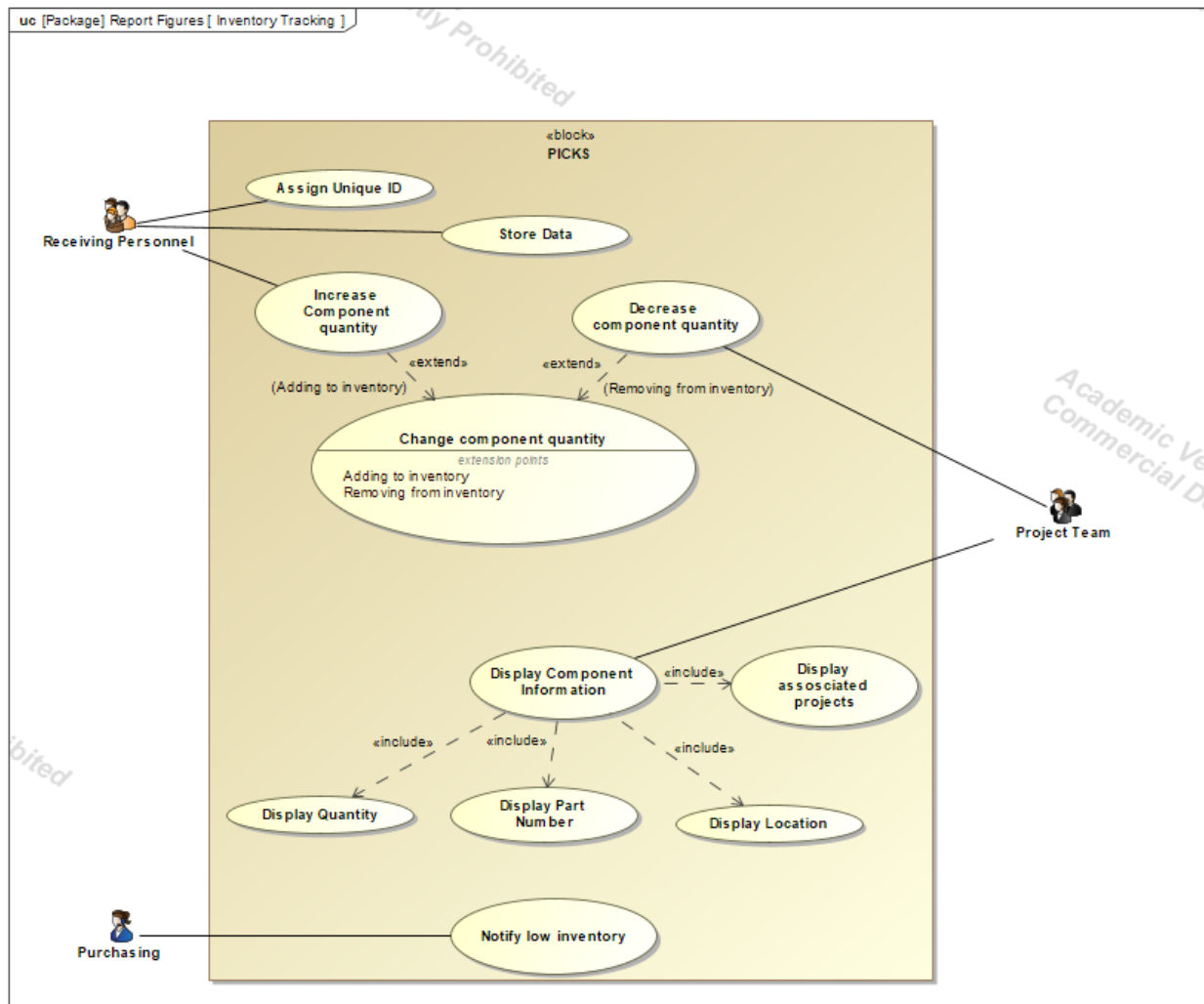


Figure 10 Inventory Tracking Use Case

For the build use case, Figure 11 Build Use Case, the builder initiates that start of a build. The system assigns a unique ID, the BOM is loaded, and a location is assigned. Material is issued to the batch. If the wrong material is scanned or there is not enough material, the builder is notified. For each correctly issued material, the unique ID for the issued material is stored in the master record. During the build, additional data may be added, such as inspections or observations per the needs of the project. Other project personnel can check the build status via the system to determine the state of the build.

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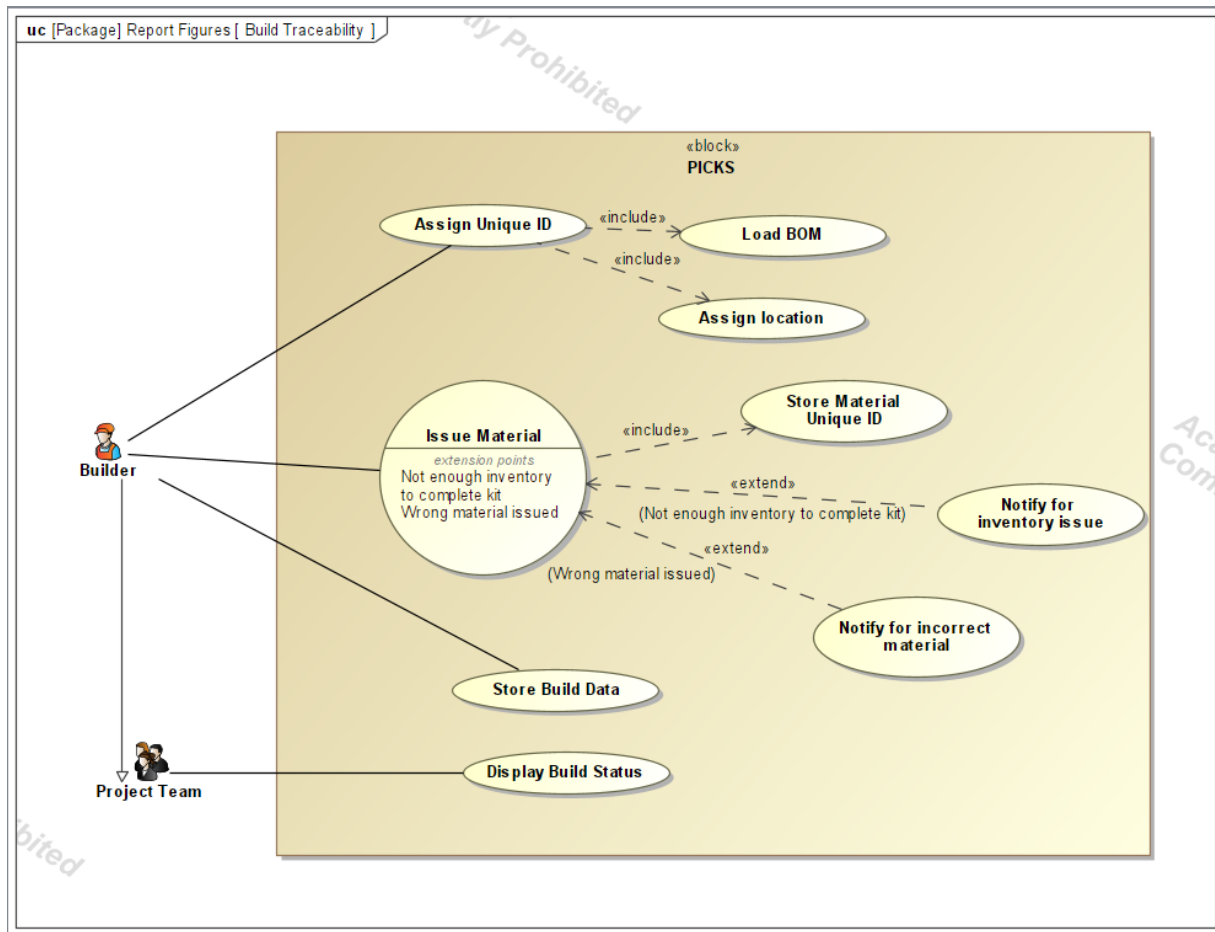


Figure 11 Build Use Case

In the kitting use case, Figure 12 Kitting Use Case, the kitting personnel initiates kitting. The unique kit ID is assigned and the BOM is loaded. The BOM indicates which material is to be pulled. Each material is issued to the kitted unique ID. The issued material unique id is stored per kitted unique ID. If there is not enough inventory or the incorrect material is used, the system will notify the personnel. The kit progress can be tracked via unique ID. This allows the user to track location, status, materials.

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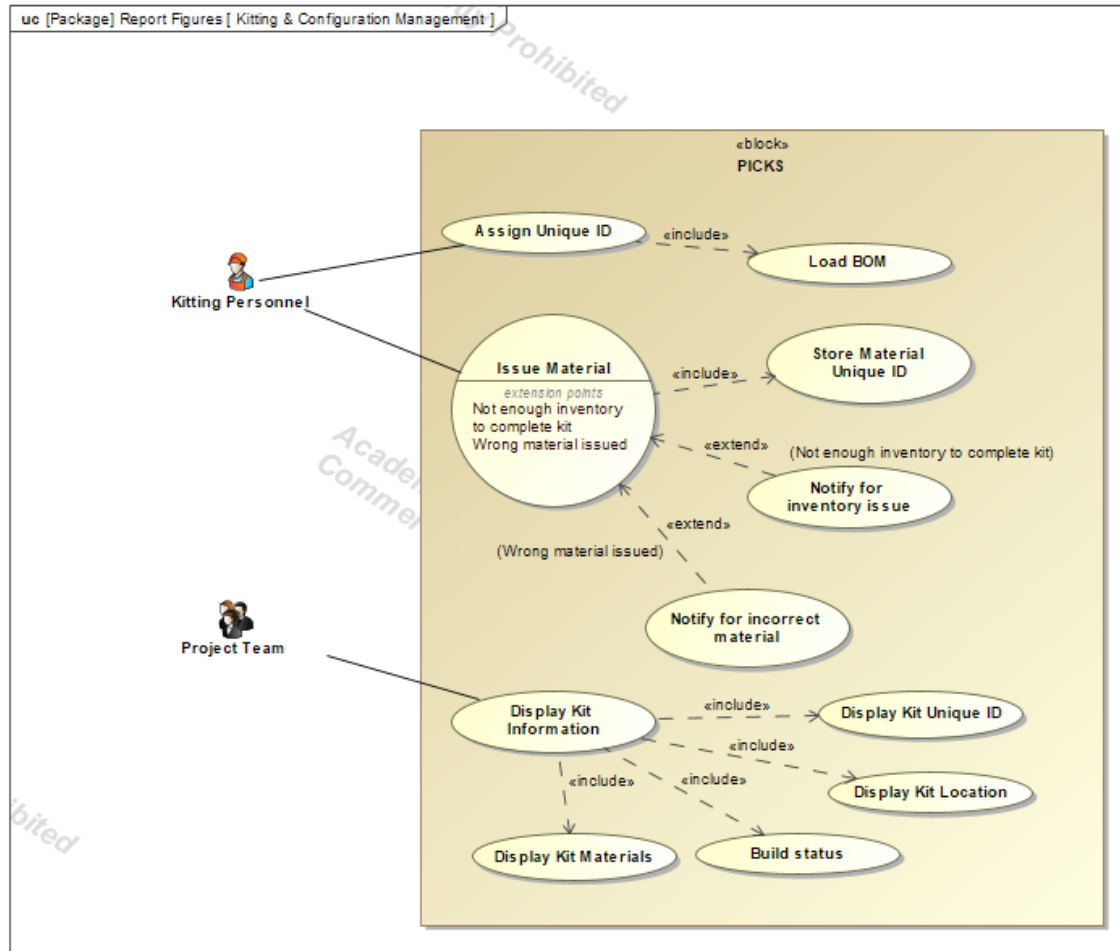


Figure 12 Kitting Use Case

In the component provenance use case, Figure 13 Component Provenance Use Case, origin data or subcomponent origin information is received, depending on the source of the information, the receiving personnel or project personnel as applicable will display the component information and add the origin data to the component information record. Assigning component origin information will include the behaviors of adding internal or external sourced origin information, adding subcomponent data, storing the data, and checking then notifying personnel or any data errors.

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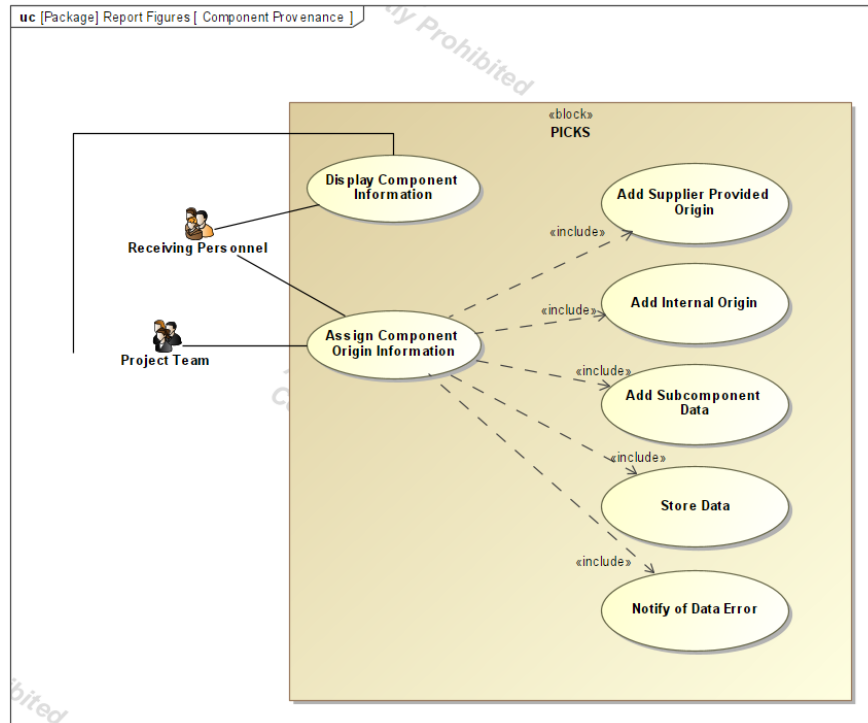


Figure 13 Component Provenance Use Case

For the track assets use case, Figure 14 Track Assets Use Case, assets IDs are created for each unique asset. The asset location is tracked by updating the location of each asset ID. When the project team needs to find the asset, it can be searched by asset ID.

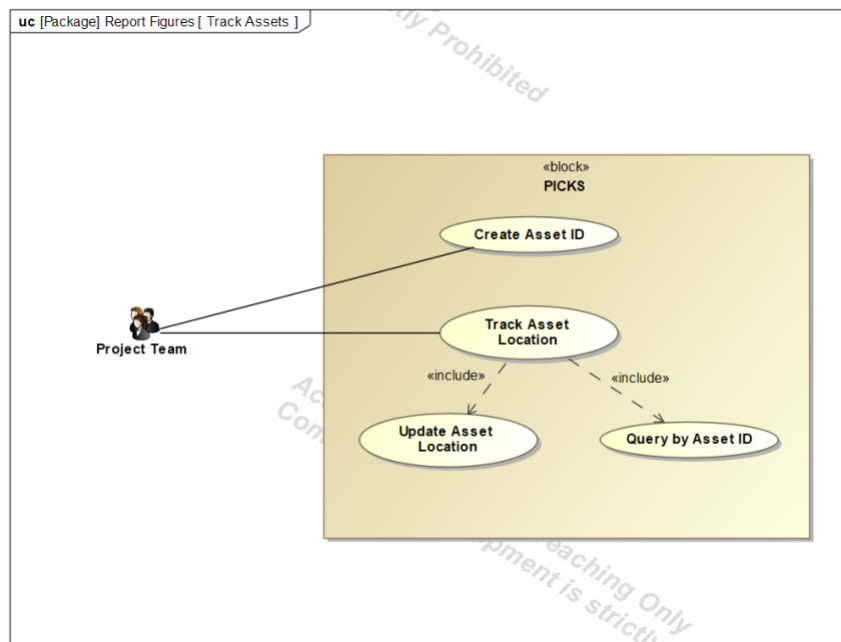


Figure 14 Track Assets Use Case

During the solution generation phase, it was announced that state of Georgia would be implementing an ERP. This ERP would most likely use a Oracle and/or work day.

6. System Architecture*

Architecture Framework and Stakeholder Organization

PICKS is intended to support stakeholder groups within the Georgia Tech Research Institute (GTRI). The architecture for PICKS is organized around the needs of different stakeholder groups. The system model separates responsibilities into distinct systems so that operational, enterprise, and compliance concerns are clearly addressed. These stakeholder groups represent the primary operational, enterprise, and compliance roles within the GTRI environment.

The primary stakeholders identified in the OV-1 include procurement and logistics teams, laboratory engineers and technicians, project and enterprise leadership, governance and cybersecurity personnel, and external compliance stakeholders such as government auditors. These stakeholders represent the operational, enterprise, and compliance roles that interact with the system.

Operational stakeholders (procurement and logistics teams, laboratory engineers, and technicians) focus on material flow, receiving efficiency, inventory accuracy, and laboratory execution performance. These concerns are addressed through the Logistics System and the Lab System, which define operational responsibilities and measurable performance factors such as receiving cycle time and execution accuracy.

Enterprise planning stakeholders (project and enterprise leadership) focus on forecast accuracy, configuration control, and coordination across domains. These concerns are addressed through the Enterprise Management System, which supports planning activities and configuration management across the organization.

Governance stakeholders (system administrators, compliance policy owners, and cybersecurity personnel) focus on meeting Cybersecurity Maturity Model Certification 2.0 requirements. They are concerned with auditability, traceability, system availability, and secure access control. These concerns are addressed through the Governance System, which includes compliance management, audit tracking, security enforcement, and authentication controls.

Executive stakeholders (project and enterprise leadership) focus on overall system performance, reliability, and the ability to demonstrate compliance. The top-level system model provides a clear representation of system boundaries and key integration points.

Organizing the architecture around stakeholder concerns ensures that operational effectiveness and compliance requirements are clearly represented and measurable within the model. The summarized architecture can be found in Figure 15 Part Architectur. Each part is discussed further in later sections.

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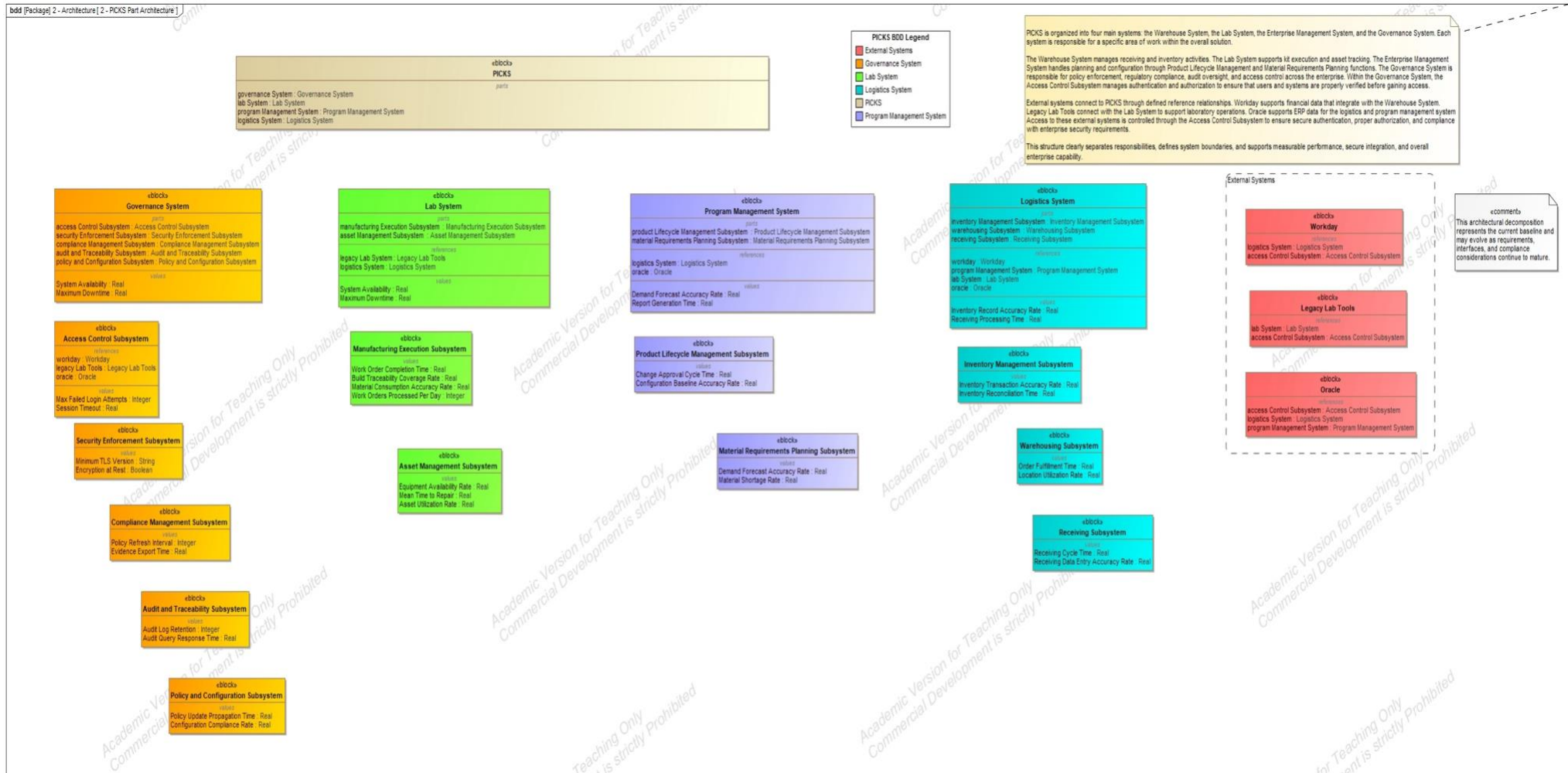


Figure 15 Part Architecture

System of Interest

PICKS is the system of interest, see Figure 16 System of Interest. It defines the system boundary and organizes all functional and managerial responsibilities within a single enterprise solution. The PICKS architecture model illustrates the main systems, their responsibilities, and their connections to external systems.

PICKS is a complex multi-mission enterprise system that integrates logistics operations, laboratory execution, enterprise planning, and governance oversight within a unified architecture.

The architecture is organized to address stakeholder concerns. Operational stakeholders are concerned with receiving and execution performance. Enterprise stakeholders are concerned with planning and configuration accuracy. Governance stakeholders are concerned with compliance, auditability, security, and availability. The model reflects these concerns by separating responsibilities into distinct systems.

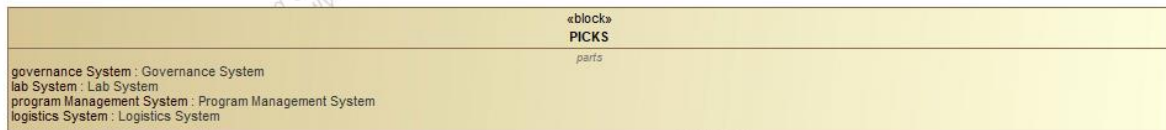


Figure 16 System of Interest

Logistics System

The purpose of the Logistics System is to manage receiving, inventory control, and warehousing operations, see Figure 17 Logistics System. It handles material flow and maintains accurate inventory records.

The system includes subsystems that support receiving, inventory management, and warehousing activities. Its behavior includes receiving materials, updating transactions, and adjusting inventory records. Performance is measured using receiving processing time and inventory record accuracy rate. These measures are important because they directly affect how efficiently materials move through the system and how accurately inventory is maintained.

Data inputs include procurement and vendor data from Oracle ERP, enterprise financial data from Workday, and internal inventory transactions. These inputs are essential because inaccurate procurement, financial, or inventory data would reduce operational performance and impact material availability.

The primary external interfaces are between Oracle, Workday, and the Logistics System. These interfaces are critical because procurement, vendor, inventory, and financial information directly affect inventory accuracy, receiving performance, and operational decision-making.

The Logistics System also works with the Program Management System and the Lab System. The Program Management System supports planning and coordination activities, while the Lab System supports laboratory execution and operational needs. These connections help ensure that inventory,

planning, and laboratory operations remain aligned. Coordination between these systems helps ensure that materials are available when needed and that operations continue without delays.

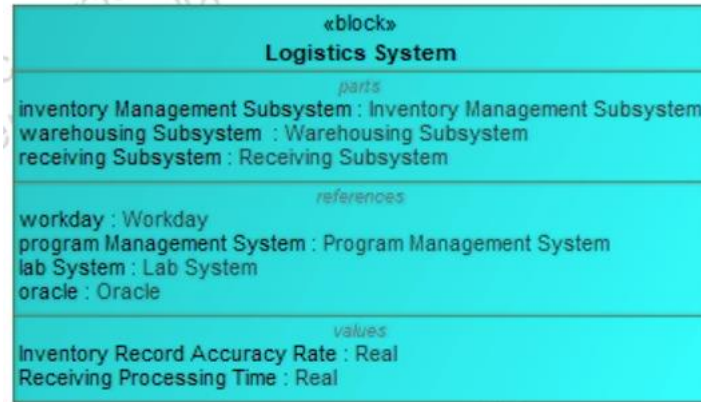


Figure 17 Logistics System

Lab System

The purpose of the Lab System, see Figure 18 Lab System is to support kit execution and asset management in laboratory operations. The system includes manufacturing execution and asset tracking subsystems. Its behavior includes kit execution activities and operational tracking. Performance is measured using execution accuracy and system availability. These measures are important because unreliable execution would reduce output quality and consistency.

Data inputs include laboratory execution data from Legacy Lab Tools and internal operational updates. Accurate laboratory data is important to ensure traceability and reliable execution.

The primary interface is Legacy Lab Tools to Lab System. This interface is important because laboratory data directly affects operational accuracy and traceability. The Lab System also works with the Logistics System. This connection helps ensure that the materials needed for kit execution are available when required and supports uninterrupted laboratory operations.

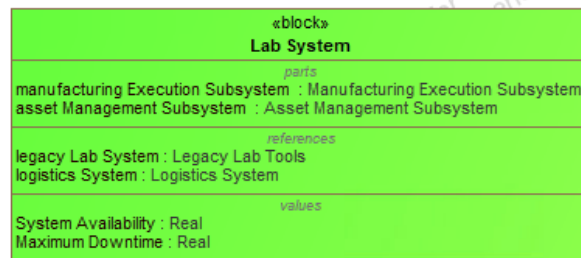


Figure 18 Lab System

Program Management System

The purpose of the Program Management System is to support planning, forecasting, and configuration management across the enterprise.

The system includes product lifecycle management and material requirements planning subsystems, see Figure 19 Program Management System. Its behavior includes forecasting, planning updates, reporting, and configuration control activities. Performance is measured using demand forecast accuracy rate and report generation time. These measures are important because inaccurate forecasts or delayed reporting would reduce operational efficiency and affect enterprise decision-making.

Data inputs include demand forecasts, configuration data, procurement information from Oracle, and operational metrics from other systems. These inputs are needed to support planning, coordination, and enterprise-level decision-making.

The primary external interface is between Oracle and the Program Management System. This interface is important because procurement and enterprise data directly affect planning activities, forecasting accuracy, and configuration management decisions.

The Program Management System also works with the Logistics System. This connection helps keep planning activities aligned with inventory operations and material availability. Coordination between these systems helps ensure that forecasts reflect actual inventory conditions and supports effective resource planning across the enterprise.

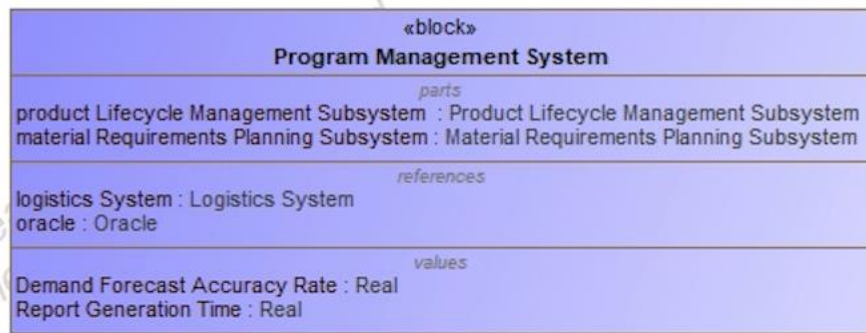


Figure 19 Program Management System

Governance System

The purpose of the Governance System is to provide compliance oversight, audit tracking, availability management, security enforcement, and access control.

The system includes subsystems for compliance management, audit tracking, security enforcement, and access control, see Figure 20 Governance System. Its behavior includes enforcing policies, verifying authentication, checking authorization, recording audit activity, and monitoring system availability. Performance is measured using audit response time, system availability, and maximum allowable downtime. These measures are important because regulatory compliance and continuous operation depend on them.

Data inputs include authentication credentials, audit records, policy definitions, and system performance metrics. The primary external interfaces are between the Access Control Subsystem and the external systems. These interfaces ensure authentication and authorization protect procurement data, laboratory data, and enterprise planning information. Failure in access control would affect security and compliance.

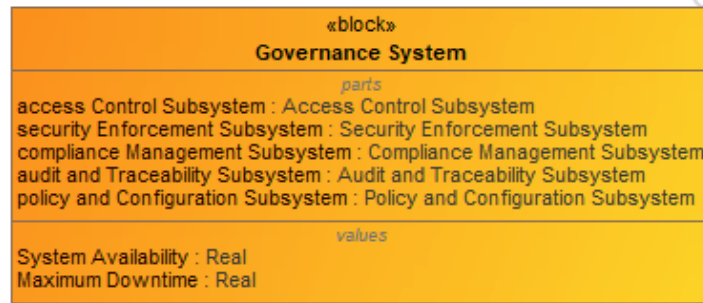


Figure 20 Governance System

Scope

The system of interest, PICKS, is a complex multi-mission enterprise system. It is not a narrowly defined system with a single mission. Instead, it supports multiple coordinated functions within one integrated solution.

PICKS combines logistics operations, laboratory execution, enterprise planning, and governance oversight within a shared system boundary, see Figure 21 External Systems. These functions work together to support and demonstrate compliance with Cybersecurity Maturity Model Certification 2.0 (CMMC 2.0) requirements.

The system is not a family of independent systems and is not a loosely connected collaborative system. All internal systems operate under centralized governance and share common compliance objectives. They function as integrated parts of one coordinated architecture.

External Systems such as Oracle, Workday, and Legacy Lab Tools connect to PICKS through defined interfaces. Oracle serves as the primary enterprise resource planning system and may be implemented using Oracle NetSuite or Oracle Fusion based on future evaluation and organizational requirements. Workday provides enterprise financial data, while Legacy Lab Tools provide laboratory execution data. These systems are outside the system boundary and are not owned or controlled by PICKS. However, their data directly affects operations, planning, and compliance activities.

The scope includes securing, monitoring, and managing these integration points to ensure traceability, data integrity, and regulatory alignment. Access to external systems is governed through the Access Control Subsystem to support authentication, authorization, auditability, and compliance requirements.

The overall effort includes defining the system architecture, implementing cybersecurity controls, enforcing access management, enabling audit evidence generation, maintaining availability, and ensuring that compliance can be demonstrated across all operational domains.

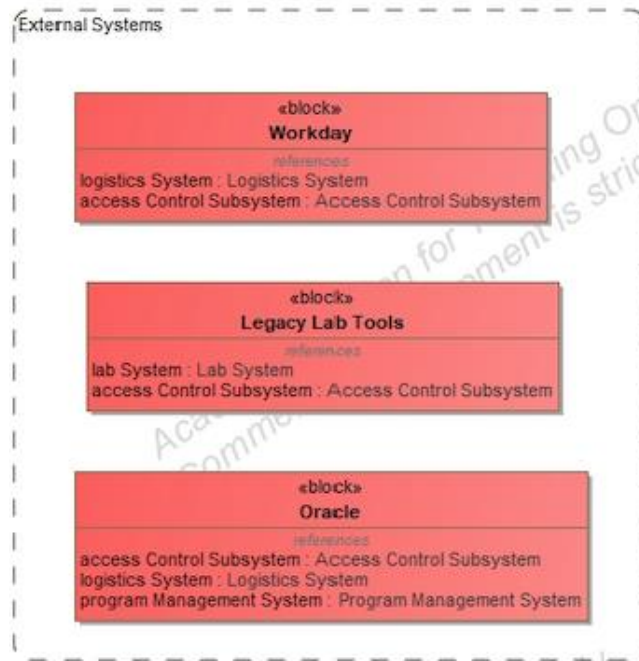


Figure 21 External Systems

Quality Attributes

The development of the system solution is driven by the need to support and demonstrate compliance with Cybersecurity Maturity Model Certification 2.0 (CMMC 2.0) requirements. The quality attributes are ranked in order of importance as Security, Traceability, Availability, Performance, and Adaptability.

Security is the highest priority because the system exists to enforce cybersecurity controls across enterprise operations. Authentication and authorization are managed through the Access Control Subsystem to ensure that only approved users and systems can access sensitive information. Protecting procurement data, laboratory data, and enterprise planning information is essential to maintain regulatory compliance and system integrity.

Traceability is critical because compliance requires verifiable evidence of control implementation and activity monitoring. The Governance System supports audit tracking, policy enforcement, and activity logging to ensure that actions can be traced, reviewed, and validated. Without traceability, compliance cannot be demonstrated or sustained.

Availability is necessary to maintain continuous oversight and monitoring. Compliance functions such as audit tracking, security enforcement, and access control must remain operational to ensure ongoing adherence to cybersecurity requirements. Defined availability targets and downtime limits support operational stability.

Performance ensures that audit response times, configuration accuracy, and operational data integrity remain within acceptable thresholds. Inaccurate or delayed reporting could weaken compliance posture and reduce confidence in system controls.

Adaptability allows the system to adjust as cybersecurity standards evolve. The separation of systems within the architecture allows updates to governance controls, access mechanisms, and compliance processes without disrupting operational systems.

7. Analysis Plan & Certification*

The development of an integrated Modeling & Simulation (M&S) Analysis Plan for PICKS centers around the main driver of the design effort: How can GTRI unify fragmented inventory, kitting, build execution, and compliance workflows into a single enterprise system that is accurate, traceable, secure, and operationally scalable across eight diverse laboratories? This question reflects the multi-mission, multi-lab nature of the system-of-interest and the need to reconcile operational efficiency with emerging compliance requirements as seen in CMMC 2.0. The M&S architecture must therefore capture not only the technical interactions between system components but also the organizational realities, legacy constraints, and stakeholder priorities that shape system behavior.

To address this, the M&S framework maps the relationships between key system inputs of inventory data, BOM structures, unique ID generation, router logic, Workday procurement data, and compliance metadata with the critical operational variables that influence system performance. These variables include traceability accuracy, build-cycle timing, kitting throughput, inventory accuracy, system availability, and the reliability of data exchanges across interfaces. Because GTRI's current processes vary widely across labs, the M&S architecture must explicitly model how these variables behave under different operational conditions, including inconsistent data quality, varying workflows, and differing compliance expectations. This resulting Verification & Validation (V&V) approach allows for core functionality to be assessed against labs-wide needs and requirements.

The V&V approach for PICKS is designed to confirm that the proposed architecture can support the operational, data-management, and compliance behaviors required across GTRI's multi-mission environment. Because PICKS integrates logistics operations, laboratory execution, enterprise planning, and governance oversight (RBAC) into a single coordinated system, the V&V framework must evaluate not only functional correctness but also interface reliability, data integrity, and compliance alignment. To accomplish this, the team developed ten pass/fail test cases that collectively assess the architecture's ability to exchange ERP purchase order and vendor data, synchronize part numbers across systems, retrieve legacy PEDYN records, update MES inventory data, enforce RBAC policies, capture audit logs, detect duplicate records, handle exceptions, and maintain end-to-end traceability. Each test case is modeled as a simple decision-based activity flow in Cameo, where successful outcomes advance toward a pass verdict and any failure in data retrieval, mapping, interface availability, or downstream usability results in a fail verdict. Together, these ten test cases provide a structured, architecture-level verification mechanism that evaluates whether PICKS can support the required operational workflows before a full implementation path is selected.

The V&V plan is grounded in a modeling and simulation (M&S) framework that captures the relationships between system inputs, including inventory data, BOM structures, unique ID generation, router logic, Workday procurement data, and compliance metadata with the operational variables that drive system performance. These variables include traceability accuracy, inventory record correctness, kitting throughput, build-cycle timing, interface reliability, and system availability. Because GTRI's

current processes vary significantly across laboratories, the V&V framework must evaluate how the architecture behaves under realistic conditions, including inconsistent data quality, asynchronous workflows, and varying compliance expectations. The Logistics System, Program Management System, Lab System, and Governance System each introduce unique dependencies and performance drivers, requiring verification activities that assess how external system data (e.g., Oracle procurement records, Workday financial data, legacy PEDYN entries) influences receiving performance, configuration control, assembly tracking, and auditability. Interface-level verification is especially critical because PICKS unifies functions traditionally distributed across ERP, MES, and PLM systems; therefore, the V&V plan includes explicit tests for data exchange, mapping accuracy, synchronization, and fallback behavior during outages or degraded operations.

The V&V plan employs a combination of discrete-event simulation, workflow modeling, and data-flow analysis to evaluate how the architecture responds to operational disruptions such as incorrect scans, insufficient inventory, mismatched BOM versions, or interface failures. Discrete-event simulation supports evaluation of material movement, kitting sequences, and build operations, enabling identification of bottlenecks and quantification of throughput. Workflow modeling validates the logic behind receiving, BOM loading, material issuance, compliance gates, and status updates, while data-flow simulation tests how information propagates across interfaces under varying loads and network conditions. Governance and security verification is also incorporated to ensure that authentication, authorization, audit logging, and tamper-evident storage behaviors meet CMMC 2.0 expectations. As requirements evolve and stakeholder feedback is incorporated, the V&V plan will be iteratively updated to ensure alignment with operational needs, compliance requirements, and system-of-systems boundaries. The PICKS team anticipates iterative updates to University System of Georgia mandated systems-of-record that PICKS will need to interface with in order to be successful. Ultimately, the V&V framework serves as both a design validation tool and a decision-support mechanism for future make-versus-buy evaluations, ensuring that PICKS can be certified as operationally effective, reliable, secure, and ready for enterprise adoption.

7.1. Test Cases

To be finalized in 6104 post presentation, finalizing test cases for the final integrated system.

To support verification of the proposed PICKS architecture, the team developed a set of ten pass/no-pass test cases covering the major functions needed for integration, inventory execution, governance, and traceability. Collectively, the test cases evaluate whether the architecture can exchange ERP purchase order and vendor data, synchronize part numbers across systems, access PEDYN legacy records, update MES inventory records, enforce RBAC policies, capture audit logs, handle exceptions, detect duplicate records, and support end-to-end traceability. Each test case was modeled as a simple activity flow in Cameo, where successful decision points proceed toward a pass verdict and unsuccessful outcomes route to a fail verdict. This approach provides a lightweight verification structure for assessing whether the proposed PICKS architecture can support the required operational and data-management behaviors before a full implementation is selected.

Test Case 1: Verify ERP Purchase Order Data Exchange (Figure 22) evaluates whether purchase order data can be successfully requested from the ERP system and made available for downstream MES or inventory workflows. The test begins by confirming that the ERP interface is available, then verifies that the integration layer can send a purchase order lookup request, the ERP can locate the requested PO record, and the integration layer can receive the returned purchase order data. The test also checks that required PO data fields are complete and correctly formatted before confirming that the validated data can be used by downstream workflows. A pass verdict is reached only if each step completes successfully; any failure in interface availability, request transmission, record retrieval, data completeness, or downstream usability results in a fail verdict.

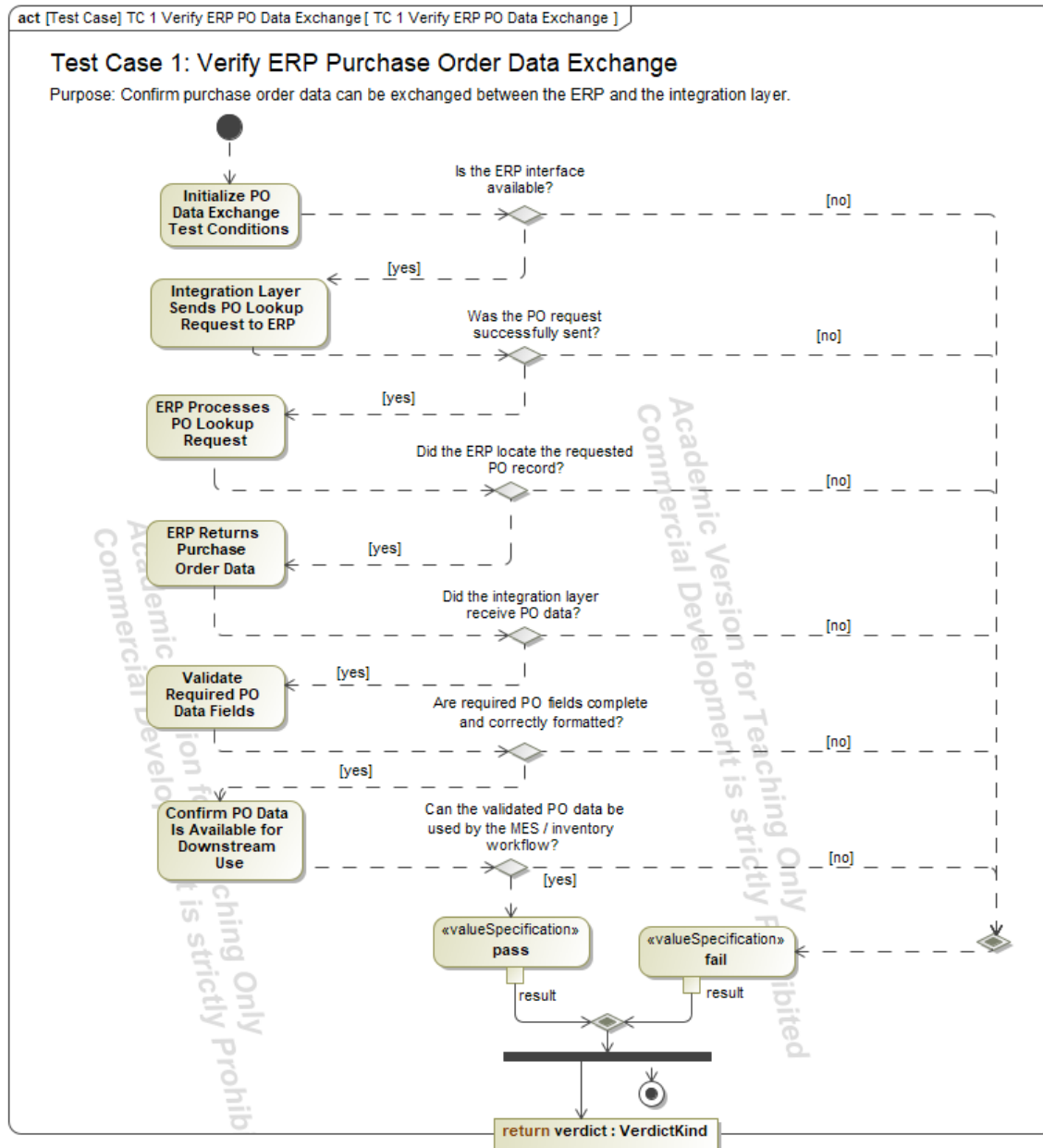


Figure 22. Test Case 1: Verify ERP Purchase Order Data Exchange

Test Case 2: Verify Vendor Data Mapping (Figure 23) evaluates whether vendor information can be correctly retrieved, mapped, and prepared for use across the integrated PICKS architecture. The test begins by confirming that a sample vendor record is available, then verifies that vendor data can be retrieved from the ERP system and processed through the integration layer using defined mapping rules. The mapped source and target fields are then compared to confirm that vendor identifiers, names, and related data elements are translated correctly between systems. The test reaches a pass verdict only if the mapped vendor record matches the expected target structure and can be used downstream by the integration layer or MES workflow. Any failure in record availability, retrieval, mapping, field comparison, or downstream usability results in a fail verdict.

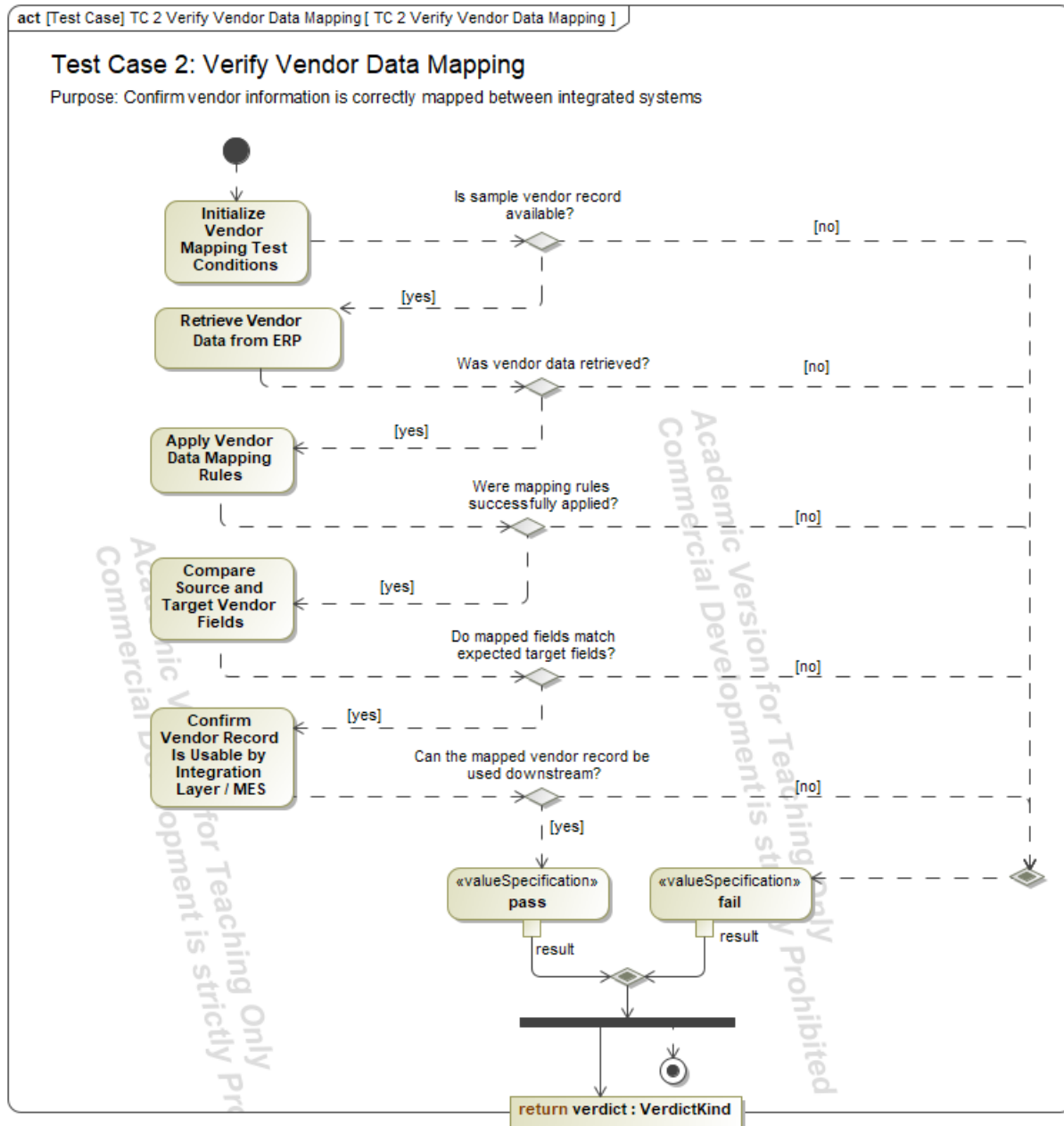


Figure 23. Test Case 2: Verify Vendor Data Mapping

Test Case 3: Verify Part Number Synchronization (Figure 24) evaluates whether part numbers remain consistent across the ERP/source system, MES candidate system, and PEDYN legacy records. The test begins by confirming that a test part number is available in the source system, then verifies that the part number can be retrieved and processed through the integration layer using the appropriate mapping rules. The synchronized part number is then checked against the MES candidate system and compared with the corresponding PEDYN legacy record to ensure there are no mismatches across systems. A pass verdict is reached only if the part number is successfully retrieved, mapped, synchronized, and confirmed to be consistent across ERP, MES, and PEDYN. Any failure in retrieval, mapping, synchronization, or cross-system consistency results in a fail verdict.

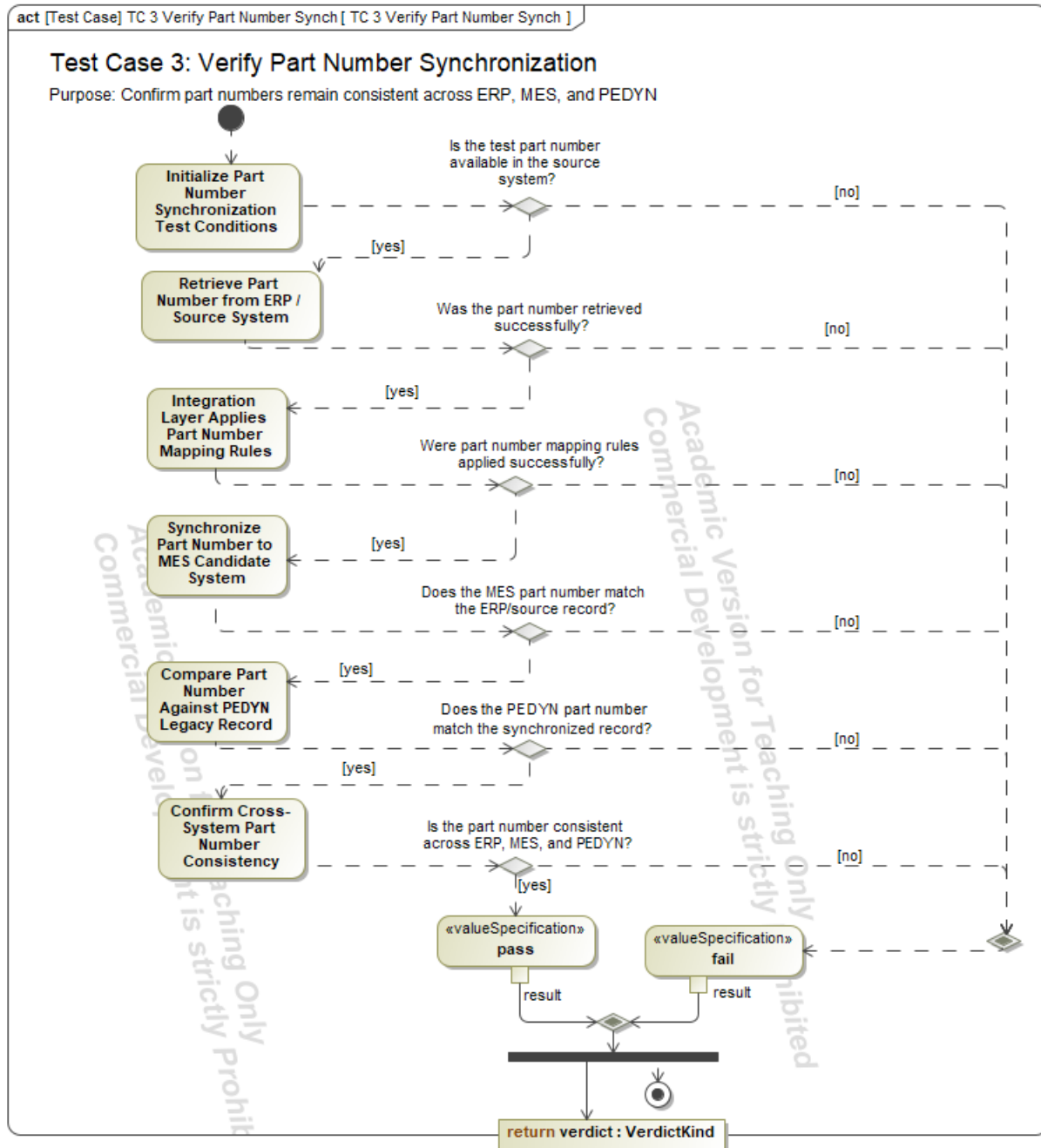


Figure 24. Test Case 3: Verify Part Number Synchronization

Additional test cases are in development. The additional test cases will be included in the final report.

8. System Design

The architecture presented in this document identifies the major systems, interfaces, and responsibilities required to support the PICKS solution. Some technology selections remain under evaluation, including the specific Oracle ERP implementation and the Manufacturing Execution

System (MES) solution. These selections will affect how systems exchange data and integrate with one another. As additional requirements are gathered and analyzed, the detailed system design will be refined and documented in the final project report.

9. Final System Solution

To be completed during ASE 6104, post presentation. Finalizing artifacts and proposals with sponsor.

10. Summary

To be completed during ASE 6104, post presentation. To be completed at the completion of the project.

11. References*

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